

## Digital Egypt Pioneers Initiative

AI & Data Science Track

# Sales Forecasting and Demand Prediction

## Project Documentation

Project Planning & Management | Literature Review | Requirements Gathering | System Analysis & Design

### Implementation Scope Note

The current implementation is planned as a Streamlit dashboard supported by a machine learning forecasting pipeline. It does not include a physical database, REST API/backend service, API testing, OOP-based class implementation, state diagram, or class diagram in the current phase.

*Prepared for Graduation Project Documentation*

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# 01

## Project Planning & Management

Sales Forecasting and Demand Prediction

## 1. Project Planning & Management

### 1. Project Proposal

#### 1.1 Project Overview

The Sales Forecasting and Demand Prediction project aims to build a machine learning model that predicts future sales and product demand based on historical data. The project utilizes the Store Sales - Time Series Forecasting dataset from Kaggle, which contains real transactional data from Corporacion Favorita, a large Ecuadorian grocery retailer.

#### 1.2 Problem Statement

Businesses often struggle with inaccurate sales forecasts, leading to overstock or stockout situations that directly impact revenue and customer satisfaction. The inability to predict demand accurately results in poor inventory management, inefficient staffing, and wasted marketing budgets. This project addresses the challenge by developing a data-driven forecasting model capable of predicting future sales across multiple product families and store locations.

#### 1.3 Project Objectives

- Perform comprehensive exploratory data analysis on the Corporacion Favorita sales dataset to uncover trends, seasonality, and key patterns.
- Engineer relevant features including time-based variables, lag features, rolling statistics, and promotional indicators.
- Develop and compare multiple forecasting models (ARIMA, Prophet, XGBoost, LSTM) to identify the best-performing approach.
- Optimize the selected model using hyperparameter tuning techniques.
- Deploy the final model as an accessible Streamlit dashboard for real-time and batch forecasting.
- Establish a monitoring framework to track model performance over time and trigger retraining when needed.

#### 1.4 Project Scope

In Scope:

- Time-series analysis and forecasting of product sales across stores.
- Feature engineering using date features, promotions, oil prices, and holiday data.
- Training and evaluation of multiple ML models.
- Model deployment via Streamlit dashboard.
- Documentation of the full data science lifecycle.

Out of Scope:

- Real-time data streaming integration.
- Mobile application development.
- Integration with live POS (Point of Sale) systems.
- Customer segmentation or marketing optimization beyond forecasting.

## 1.5 Dataset Description

The project uses the Kaggle Store Sales - Time Series Forecasting competition dataset, which includes the following files:

| File                | Size (approx.) | Description                                                                           |
|---------------------|----------------|---------------------------------------------------------------------------------------|
| train.csv           | ~3M rows       | Historical sales data with date, store, product family, sales, and onpromotion fields |
| test.csv            | ~28K rows      | Test set for forecasting future 15-day window                                         |
| stores.csv          | 54 rows        | Store metadata: city, state, type, cluster                                            |
| oil.csv             | ~1.2K rows     | Daily oil price data (Ecuador economy is oil-dependent)                               |
| holidays_events.csv | ~350 rows      | National and regional holidays and special events                                     |
| transactions.csv    | ~83K rows      | Number of transactions per store per day                                              |

## 1.6 Expected Outcomes

- A fully trained and evaluated sales forecasting model with documented performance metrics (RMSE, MAE, RMSLE).
- An interactive dashboard or dashboard interface enabling stakeholders to query sales predictions.
- Comprehensive documentation covering data exploration, modeling decisions, and business insights.
- A reusable ML pipeline ready for retraining with updated data.

## 2. Project Plan

### 2.1 Project Timeline & Milestones

The project follows a structured 5-milestone plan aligned with the DEPI documentation deadlines:

| Milestone                                        | Key Deliverables                                 | Start Date   | Deadline     |
|--------------------------------------------------|--------------------------------------------------|--------------|--------------|
| M1: Data Collection, Exploration & Preprocessing | EDA Report, Cleaned Dataset, Visualizations      | Mar 1, 2026  | Mar 31, 2026 |
| M2: Advanced Analysis & Feature Engineering      | Analysis Report, Feature Engineering Summary     | Apr 1, 2026  | Apr 30, 2026 |
| M3: Model Development & Optimization             | Model Evaluation Report, Model Code, Final Model | May 1, 2026  | Jun 15, 2026 |
| M4: MLOps, Deployment & Monitoring               | Deployed Model, MLOps Report, Monitoring Setup   | Jun 16, 2026 | Jul 5, 2026  |
| M5: Final Documentation & Presentation           | Final Report, Final Presentation                 | Jul 6, 2026  | Jul 17, 2026 |

## 2.2 Gantt Chart (Simplified Overview)

| Task / Month                  | Mar | Apr | May | Jun (H1) | Jun (H2) | Jul (H1) | Jul (H2) | Aug |
|-------------------------------|-----|-----|-----|----------|----------|----------|----------|-----|
| M1: Data Collection & EDA     | ■   | ■   |     |          |          |          |          |     |
| M2: Feature Engineering       |     | ■   | ■   |          |          |          |          |     |
| M3: Model Development         |     |     | ■   | ■        |          |          |          |     |
| M4: Deployment & MLOps        |     |     |     |          | ■        | ■        |          |     |
| M5: Final Docs & Presentation |     |     |     |          |          |          | ■        | ■   |

## 2.3 Deliverables Summary

| Deliverable                    | Responsible Party       | Due Date     |
|--------------------------------|-------------------------|--------------|
| EDA Report & Notebook          | Data Analyst(s)         | Mar 31, 2026 |
| Cleaned & Preprocessed Dataset | Data Engineer(s)        | Mar 31, 2026 |
| Feature Engineering Summary    | Data Analyst(s)         | Apr 30, 2026 |
| Model Evaluation Report        | ML Engineer(s)          | Jun 15, 2026 |
| Model Code (Python)            | ML Engineer(s)          | Jun 15, 2026 |
| Deployed Model (API/Dashboard) | DevOps / ML Engineer(s) | Jul 5, 2026  |
| MLOps & Monitoring Report      | DevOps Engineer(s)      | Jul 5, 2026  |
| Final Project Report           | All Team Members        | Jul 17, 2026 |
| Final Presentation (PPT)       | All Team Members        | Jul 17, 2026 |

## 2.4 Resource Allocation

- Human Resources: Team of 4-6 students with defined roles (see Section 3).
- Tools & Technologies: Python, Jupyter Notebook, Scikit-learn, XGBoost, Prophet, TensorFlow/PyTorch, Streamlit, MLflow, GitHub.
- Data Resources: Kaggle Store Sales dataset (publicly available, free to download).
- Infrastructure: Local development machines + optional cloud deployment (Google Colab / AWS Free Tier / Heroku).

## 3. Task Assignment & Roles

### 3.1 Team Structure

The team consists of [5] members with the following roles and responsibilities. Each member is responsible for their assigned tasks but collaboration is expected across all phases.

| Team Member | Role                           | Primary Responsibilities                               | Milestone(s) |
|-------------|--------------------------------|--------------------------------------------------------|--------------|
| Ahmed       | Project Manager / Data Analyst | Project coordination, final reporting                  | M1, M5       |
| Mohamed     | Data Engineer                  | Data cleaning, preprocessing, EDA, feature engineering | M1, M2       |
| Shimaa      | ML Engineer                    | Model development, training, evaluation, tuning        | M3           |
| Fatma       | MLOps / DevOps Engineer        | Model deployment, MLflow, monitoring setup             | M4           |
| Youssef     | Data Visualizer / Analyst      | Dashboards, visualizations, presentation               | M2, M5       |
| Ahmed       | ML Engineer (Support)          | Model experimentation, hyperparameter tuning           | M3, M4       |

### 3.2 Detailed Responsibilities

#### Project Manager / Data Analyst

- Oversee project timeline and ensure deadlines are met.
- Coordinate between team members and communicate with the instructor.
- Lead the EDA phase: data loading, initial exploration, and summary statistics.
- Compile the final project report and coordinate the final presentation.

#### Data Engineer

- Download and prepare the Kaggle dataset for use.
- Handle missing values, duplicates, and data inconsistencies.
- Engineer time-based features (month, week, day-of-week, seasonality flags).
- Create lag features, rolling averages, and promotional flag features.
- Merge auxiliary datasets (oil prices, holidays, transactions, store metadata).

**ML Engineer**

- Select appropriate forecasting models: SARIMA, Linear Regression, XGBoost, Random Forest.
- Implement train-validation-test splits respecting time series order.
- Train models and evaluate using RMSE, MAE, RMSLE metrics.
- Conduct hyperparameter tuning using Grid Search or Random Search.
- Document model comparison results and final model selection rationale.

**MLOps / DevOps Engineer**

- Set up MLflow for experiment tracking, model versioning, and artifact logging.
- Deploy the final model using Streamlit or Streamlit.
- Configure model performance monitoring and drift detection.
- Set up alert mechanisms for performance degradation.
- Document the deployment pipeline and retraining strategy.

**Data Visualizer / Analyst**

- Create visualizations: line plots, bar charts, heatmaps, seasonal decomposition plots.
- Build interactive dashboards using Plotly or Streamlit.
- Prepare the final presentation slides.
- Ensure all charts and visuals are clearly labeled and meaningful.

**4. Risk Assessment & Mitigation Plan****4.1 Risk Matrix**

| Risk                                                     | Likelihood | Impact | Severity | Mitigation Strategy                                                                      |
|----------------------------------------------------------|------------|--------|----------|------------------------------------------------------------------------------------------|
| Dataset quality issues (missing values, inconsistencies) | Medium     | High   | High     | Apply robust preprocessing; use multiple imputation strategies                           |
| Model underfitting / poor forecast accuracy              | Medium     | High   | High     | Test multiple model types; tune hyperparameters; add more features                       |
| Team member unavailability                               | Low        | Medium | Medium   | Cross-train team members; document all tasks clearly                                     |
| Computational resource limitations                       | Medium     | Medium | Medium   | Use Google Colab Pro / cloud platforms; optimize code efficiency                         |
| Deployment failures / technical issues                   | Low        | High   | Medium   | Test deployment locally first; use Docker containers; have backup plan (Streamlit Cloud) |
| Overfitting to training data                             | Medium     | High   | High     | Use proper time-series cross-validation; apply regularization techniques                 |
| Scope creep / feature overload                           | Medium     | Medium | Medium   | Stick to defined scope; document any changes with instructor approval                    |
| Version control conflicts                                | Low        | Low    | Low      | Use Git branching strategy; regular commits and pull request reviews                     |



## 4.2 Risk Response Strategy

### Data Quality Risks

The dataset from Kaggle is well-maintained for competition purposes, but missing values in oil prices and occasional gaps in sales data are expected. The team will apply forward-fill and interpolation strategies for time series gaps and document all preprocessing decisions.

### Technical Risks

For computational limitations, the team will leverage Google Colab with GPU acceleration for deep learning experiments (LSTM). For classical models (ARIMA, Prophet, XGBoost), local machines are sufficient.

### Timeline Risks

Weekly team check-ins will be held to monitor progress. If a milestone is at risk of being delayed, the team will reallocate tasks and notify the instructor proactively.

## 5. Key Performance Indicators (KPIs)

### 5.1 Model Performance KPIs

The following metrics will be used to evaluate the success of the forecasting model:

| KPI                            | Description                                                    | Target   | Measurement Method        |
|--------------------------------|----------------------------------------------------------------|----------|---------------------------|
| MAE (Mean Absolute Error)      | Average absolute difference between predicted and actual sales | Minimize | Validation set evaluation |
| RMSE (Root Mean Squared Error) | Penalizes large forecast errors more heavily                   | Minimize | Validation set evaluation |
| R-Squared ( $R^2$ )            | Proportion of variance explained by the model                  | > 0.80   | Validation set evaluation |
| Model Training Time            | Time to train the final model                                  | < 20 min | Logged via MLflow         |

## 5.2 Project Delivery KPIs

| KPI                        | Target                                              | Measurement Method                       |
|----------------------------|-----------------------------------------------------|------------------------------------------|
| Milestone Completion Rate  | 100% of milestones delivered on time                | GitHub commit history & submission dates |
| Documentation Completeness | All required documents submitted per guidelines     | Checklist review by Project Manager      |
| Code Quality               | Well-commented, modular, PEP8-compliant Python code | Code review + pylint score > 7/10        |
| Team Task Coverage         | Each member contributes to at least 2 milestones    | GitHub contribution graph                |
| Deployment Uptime          | Deployed model accessible and functional            | Manual testing at submission             |

## 5.3 Business Impact KPIs

- **Forecast Accuracy Improvement:** The model should outperform a naive baseline (last year's same-period sales) by at least 20% in terms of MAE.
- **Actionable Insights:** The EDA and feature analysis should identify at least 3 key drivers of sales variation (e.g., promotions, holidays, oil price).
- **Stakeholder Usability:** The deployed dashboard/API should allow a non-technical user to generate forecasts with minimal instructions.

# 02

## Literature Review

Sales Forecasting and Demand Prediction

## 2. Literature Review

### 1. Introduction

Accurate sales forecasting is a cornerstone of modern retail and supply-chain management. The ability to predict future demand with high precision enables businesses to optimize inventory levels, allocate resources efficiently, and make informed strategic decisions. Inaccurate forecasts, by contrast, lead to either costly overstock situations or missed revenue due to stockouts.

This literature review examines the evolution of time-series forecasting methods as applied to retail sales prediction, with a specific focus on the techniques and tools relevant to our project: the Store Sales - Time Series Forecasting dataset from Corporacion Favorita (Kaggle). The review covers classical statistical methods, modern machine learning approaches, deep learning architectures, and hybrid strategies, culminating in a comparative summary that informs our modeling choices.

The review is organized as follows: Section 2 provides background on time-series forecasting fundamentals. Section 3 surveys classical statistical models. Section 4 covers machine learning-based approaches. Section 5 examines deep learning methods. Section 6 discusses feature engineering best practices. Section 7 reviews MLOps and deployment considerations. Section 8 presents a comparative analysis. Section 9 concludes with key takeaways.

### 2. Background: Time-Series Forecasting in Retail

#### 2.1 What is Time-Series Forecasting?

A time series is a sequence of data points indexed in chronological order. Time-series forecasting involves using historical observations to predict future values. In the retail context, the target variable is typically daily or weekly sales volume per product category and store location.

Forecasting models must account for several key temporal patterns:

- **Trend:** The long-term upward or downward movement of sales over time (e.g., growing brand popularity).
- **Seasonality:** Recurring patterns tied to calendar cycles (e.g., higher sales during holidays, weekends, or specific months).
- **Cyclicity:** Longer-term fluctuations often tied to economic conditions (e.g., oil price dependency in Ecuador).
- **Irregularity / Noise:** Random variation that cannot be systematically explained.

#### 2.2 The Corporacion Favorita Problem

The Kaggle Store Sales dataset presents a multi-variate, multi-step forecasting challenge. The goal is to predict 15 days of future sales across 54 stores and 33 product families, using features including:

- Historical sales per store and product family.
- Promotional flags indicating whether a product was on promotion on a given day.
- Ecuador national holidays and regional events that significantly impact consumer behavior.
- Daily oil prices (Ecuador's economy is heavily oil-dependent, which influences consumer spending power).
- Number of daily transactions per store (a proxy for foot traffic).

This richness of contextual features makes the problem well-suited for both classical and modern ML approaches, allowing for extensive feature engineering and multivariate modeling.

## 3. Classical Statistical Forecasting Methods

### 3.1 ARIMA and SARIMA

The AutoRegressive Integrated Moving Average (ARIMA) model, introduced by Box and Jenkins (1970), remains one of the most widely used univariate forecasting methods. ARIMA combines three components: autoregression (AR), differencing for stationarity (I), and moving average (MA), parameterized as ARIMA(p, d, q).

The Seasonal ARIMA (SARIMA) extension adds seasonal counterparts for each component, making it suitable for data with strong periodic patterns. SARIMA is parameterized as  $ARIMA(p,d,q)(P,D,Q)[s]$ , where  $s$  is the seasonal period (e.g., 7 for weekly data, 12 for monthly data).

Strengths and limitations in the retail context:

- Strengths: Well-understood mathematically; interpretable coefficients; effective for stationary or easily-differenced series.
- Limitations: Assumes linear relationships; struggles with multiple seasonalities; requires separate models per store/product combination - computationally expensive at scale.

[1]

Box, G.E.P., Jenkins, G.M., Reinsel, G.C., & Ljung, G.M. (2015 (original 1970)). "Time Series Analysis: Forecasting and Control (5th ed.)." Wiley. ISBN: 978-1-118-67502-1.  
**Relevance:** Foundational reference for ARIMA methodology applied in retail and demand forecasting.

### 3.2 Exponential Smoothing (ETS)

Exponential Smoothing State Space (ETS) models assign exponentially decreasing weights to past observations, giving more importance to recent data. The Holt-Winters method extends simple exponential smoothing to capture both trend and seasonality through additive or multiplicative components.

ETS models are computationally efficient and often competitive with more complex methods on shorter horizons. However, like ARIMA, they are univariate and cannot directly incorporate external regressors such as promotions or holidays.

[2]

Hyndman, R.J. & Athanasopoulos, G. (2021). "Forecasting: Principles and Practice (3rd ed.)." OTexts. Available at: [otexts.com/fpp3](https://otexts.com/fpp3).  
**Relevance:** Comprehensive open-access textbook covering ETS, ARIMA, and modern forecasting methods with R implementations.

### 3.3 Prophet (Facebook / Meta)

Prophet, developed by Taylor and Letham at Facebook (2018), is a decomposable additive model of the form:  $y(t) = \text{trend}(t) + \text{seasonality}(t) + \text{holidays}(t) + \text{error}(t)$ . It was specifically designed for business time series with the following characteristics:

- Multiple seasonal periods (daily, weekly, yearly).
- Known irregular holidays and special events.
- Missing data and outliers are handled gracefully.
- Non-linear trend changes (changepoints) detected automatically.

Prophet is particularly relevant to our project because it natively supports holiday regressors - directly modeling the Ecuadorian holiday calendar included in the Kaggle dataset. It is implemented in Python (and R) and requires minimal hyperparameter tuning, making it accessible for rapid prototyping.

[3]

Taylor, S.J. & Letham, B. (2018). "Forecasting at Scale." *The American Statistician*, 72(1), pp. 37-45. DOI: 10.1080/00031305.2017.1380080.  
 Relevance: Original Prophet paper; directly relevant as we plan to use Prophet as one of our baseline forecasting models.

## 4. Machine Learning-Based Forecasting Methods

### 4.1 Gradient Boosting: XGBoost and LightGBM

Gradient Boosting Decision Trees (GBDT) have become the dominant approach in structured/tabular forecasting competitions. XGBoost (Extreme Gradient Boosting), introduced by Chen and Guestrin (2016), and LightGBM (Microsoft, 2017) are the two most widely used implementations.

In time-series forecasting, these models do not natively understand temporal ordering. Instead, the time dimension is encoded through feature engineering - creating lag features (e.g., sales\_lag\_7, sales\_lag\_14), rolling statistics (e.g., rolling\_mean\_28), and calendar features (day\_of\_week, month, is\_holiday). This transforms the problem into a standard supervised regression task.

Key advantages for our Kaggle problem:

- Handles mixed feature types: numerical (oil price, lag sales) and categorical (product family, store type) natively with label encoding.
- Captures non-linear interactions between promotions, holidays, and sales without explicit specification.
- Scales well to large datasets; can model all stores and product families jointly with store/product embedding features.
- Consistently top-performing in Kaggle tabular competitions including previous sales forecasting challenges.

[4]

Chen, T. & Guestrin, C. (2016). "XGBoost: A Scalable Tree Boosting System." *Proceedings of KDD 2016*, pp. 785-794. DOI: 10.1145/2939672.2939785.  
 Relevance: Original XGBoost paper; our primary ML model candidate given its dominance in Kaggle forecasting competitions.

[5]

Ke, G., Meng, Q., Finley, T., et al. (2017). "LightGBM: A Highly Efficient Gradient Boosting Decision Tree." *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30.  
 Relevance: LightGBM alternative to XGBoost; faster training on large datasets, relevant for our multi-store forecasting scenario.

### 4.2 Random Forest for Time Series

Random Forest (Breiman, 2001) is an ensemble of decision trees trained on bootstrap samples with random feature subsets. For time-series forecasting, it is applied similarly to XGBoost - via lag and calendar feature engineering.

While generally outperformed by gradient boosting on tabular data, Random Forest serves as an important baseline model due to its robustness to overfitting and ease of interpretation. Feature importance scores from Random Forest can inform the feature engineering process for more complex models.

[6]

Breiman, L. (2001). "Random Forests." *Machine Learning*, 45(1), pp. 5-32. DOI: 10.1023/A:1010933404324.  
 Relevance: Foundational Random Forest paper; used as a baseline comparison model in our forecasting pipeline.

### 4.3 Support Vector Regression (SVR)

Support Vector Regression extends the SVM classifier to regression tasks. For time-series forecasting, SVR has shown competitive performance on short-horizon univariate tasks. However, its computational complexity scales poorly with dataset size ( $O(n^2)$  to  $O(n^3)$ ), making it impractical for the Corporacion Favorita dataset with millions of observations. We include it in this review for completeness but do not prioritize it for implementation.

## 5. Deep Learning Approaches

### 5.1 LSTM (Long Short-Term Memory Networks)

Long Short-Term Memory networks (Hochreiter & Schmidhuber, 1997) are a type of Recurrent Neural Network (RNN) designed to capture long-range temporal dependencies. LSTMs use gating mechanisms (input gate, forget gate, output gate) to selectively retain or discard information from past time steps, addressing the vanishing gradient problem of vanilla RNNs.

In retail sales forecasting, LSTMs have demonstrated strong performance when:

- The training dataset is large enough (millions of observations) to support deep learning.
- Long-term seasonal patterns are present that shallow models miss.
- Multiple related time series (stores x product families) can be jointly modeled in a global architecture.

For our project, we plan to experiment with a basic LSTM architecture using Google Colab GPU resources. The input will be a sliding window of the past 28-56 days of sales, with categorical embeddings for store and product family.

[7]

Hochreiter, S. & Schmidhuber, J. (1997). "Long Short-Term Memory." *Neural Computation*, 9(8), pp. 1735-1780. DOI: 10.1162/neco.1997.9.8.1735.  
 Relevance: Original LSTM paper; directly relevant as LSTM is one of our planned deep learning model candidates.

### 5.2 Temporal Convolutional Networks (TCN)

Temporal Convolutional Networks apply dilated causal convolutions to time-series data, allowing the model to capture patterns at multiple temporal scales without the sequential computation bottleneck of RNNs. TCNs have been shown to match or outperform LSTMs on many time-series benchmarks while training faster due to parallelization.

[8]

Bai, S., Kolter, J.Z., & Koltun, V. (2018). "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling." *arXiv:1803.01271*.  
 Relevance: Demonstrates TCN performance vs LSTM; informs our choice of architectures to compare in the deep learning phase.

### 5.3 Transformer-Based Models (N-BEATS, TFT)

The Temporal Fusion Transformer (TFT), proposed by Lim et al. (2021), combines multi-head attention with gating mechanisms to handle both static (store metadata) and dynamic (promotional calendar, oil price) covariates. It won the M5 forecasting competition in 2020.

While TFT represents the state of the art, its implementation complexity and computational demands exceed our project scope. We document it here as a potential future improvement and to contextualize our chosen approaches within the broader research landscape.

[9]

Lim, B., Arik, S.O., Loeff, N., & Pfister, T. (2021). "Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting." *International Journal of Forecasting*, 37(4), pp. 1748-1764. DOI: 10.1016/j.ijforecast.2021.03.012. Relevance: State-of-the-art Transformer forecasting model; referenced as a future improvement direction for our project.

## 6. Feature Engineering for Sales Forecasting

### 6.1 Time-Based Features

Calendar-derived features are foundational for retail forecasting. Research consistently shows that encoding temporal context significantly improves model accuracy. Key features identified in the literature include:

- Day-of-week (0-6): Captures weekly seasonality - weekends typically see different shopping patterns than weekdays.
- Month and quarter: Captures monthly and quarterly seasonality.
- Day-of-year: Captures annual seasonality for data with multiple years of history.
- Is\_weekend, is\_holiday, days\_until\_holiday, days\_after\_holiday: Critical for the Ecuadorian holiday calendar in our dataset.
- Week-of-year: Useful for identifying recurring annual events like school seasons.

### 6.2 Lag Features and Rolling Statistics

Lag features encode past sales values as predictors for future values, directly capturing autocorrelation in the series. Rolling statistics summarize the behavior over recent windows. Both are essential for gradient boosting models that cannot natively process sequential data.

| Feature Type    | Example Features                   | Rationale                                                                  |
|-----------------|------------------------------------|----------------------------------------------------------------------------|
| Lag Features    | sales_lag_1, lag_7, lag_14, lag_28 | Direct autocorrelation; last week / last month sales are strong predictors |
| Rolling Mean    | rolling_mean_7, rolling_mean_28    | Smooths short-term noise; captures medium-term trend                       |
| Rolling Std Dev | rolling_std_7, rolling_std_28      | Captures sales volatility; important for irregular categories              |
| Rolling Max/Min | rolling_max_7, rolling_min_28      | Captures peak/trough patterns around promotions                            |
| Expanding Mean  | expanding_mean_to_date             | Historical average; useful baseline for each store-product pair            |



### 6.3 Promotion and Holiday Effects

Promotional periods and holidays are among the strongest drivers of retail sales variation. Research by Fildes et al. (2019) showed that models explicitly incorporating promotion data outperform those that do not by 15-25% in MAPE terms. For our dataset:

- The onpromotion column in train.csv directly flags promotional days per product-store pair.
- The holidays\_events.csv file allows us to create features for national holidays, regional events, and transferred holidays.
- Interaction features (e.g., is\_promotion AND is\_weekend) can capture compounding effects.

[10]

Fildes, R., Ma, S., & Kolassa, S. (2019). "Retail Forecasting: Research and Practice." *International Journal of Forecasting*, 38(4), pp. 1283-1318. DOI: 10.1016/j.ijforecast.2019.06.004.  
**Relevance:** Comprehensive review of retail forecasting; specifically addresses promotion effects - directly applicable to our dataset.

### 6.4 External Regressors: Oil Prices

The Corporacion Favorita dataset uniquely includes daily oil prices as an external regressor. Ecuador's economy is highly sensitive to oil price fluctuations, which affect consumer purchasing power. Research in macroeconomic forecasting has demonstrated that including relevant economic indicators as exogenous variables improves model accuracy for consumer goods sales.

For our project, we will engineer oil price features including: current oil price, oil price lag (7-day, 30-day), and oil price rolling trend (30-day moving average), to capture both immediate and lagged economic effects on sales.

## 7. Evaluation Metrics for Forecasting Models

Selecting appropriate evaluation metrics is critical for comparing forecasting models fairly. The literature identifies several commonly used metrics, each with different properties and use cases:

| Metric    | Formula (conceptual)                                  | Properties                                                  | Use in Our Project                             |
|-----------|-------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------|
| RMSE      | $\sqrt{\text{mean}((\text{pred} - \text{act})^2)}$    | Penalizes large errors heavily; same units as target        | Secondary metric for model comparison          |
| MAE       | $\text{mean}( \text{pred} - \text{act} )$             | Robust to outliers; easy to interpret; same units as target | Reported alongside RMSE for completeness       |
| R-Squared | $1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$ | Proportion of variance explained; 1 = perfect fit           | Used to assess overall model explanatory power |

The Kaggle competition uses RMSLE as the primary evaluation metric. The log transformation is particularly appropriate for sales data because: (1) sales values span multiple orders of magnitude across different product categories, and (2) it penalizes under-forecasting (which causes stockouts) more than over-forecasting (which causes excess inventory), reflecting real business priorities.

[11]

Hyndman, R.J. & Koehler, A.B. (2006). "Another Look at Measures of Forecast Accuracy." *International Journal of Forecasting*, 22(4), pp. 679-688. DOI: 10.1016/j.ijforecast.2006.03.001.  
**Relevance:** Seminal paper on forecast accuracy metrics; informs our choice and interpretation of RMSE, MAE, and MAPE.

## 8. MLOps and Model Deployment in Forecasting

### 8.1 Experiment Tracking with MLflow

MLflow (Zaharia et al., 2018) is an open-source platform for managing the machine learning lifecycle, including experiment tracking, model versioning, and deployment. In the context of our project, MLflow will be used to:

- Log hyperparameters, evaluation metrics (MAE, RMSE), and training duration for each model run.
- Store model artifacts (serialized model files) with versioning to enable reproducibility.
- Compare runs visually through the MLflow UI to identify the best-performing configuration.

[12]

Zaharia, M., Chen, A., Davidson, A., et al. (2018). "Accelerating the Machine Learning Lifecycle with MLflow." *IEEE Data Engineering Bulletin*, 41(4), pp. 39-45.  
**Relevance:** Original MLflow paper; our chosen experiment tracking tool for managing model versions and metrics.

### 8.2 Model Deployment with Flask / Streamlit

For production deployment of machine learning models, lightweight Python web frameworks are the standard approach. Streamlit has emerged as the preferred modern alternative to Flask due to its:

- Automatic API documentation generation via OpenAPI/Swagger.
- Asynchronous request handling for improved throughput.
- Native Pydantic data validation for request/response schemas.

Our deployment plan involves wrapping the trained forecasting model in a Streamlit endpoint that accepts store ID, product family, and date range as inputs and returns predicted sales values as JSON output.

### 8.3 Interactive Dashboards with Streamlit

Streamlit enables rapid development of interactive data applications in pure Python. For our project, a Streamlit dashboard will allow non-technical stakeholders (e.g., store managers, business analysts) to:

- Select a store and product family from dropdown menus.
- View historical sales trends with interactive Plotly charts.
- Generate and visualize sales forecasts for a configurable future horizon.
- Understand the impact of upcoming promotions and holidays on predicted demand.

### 8.4 Model Drift and Monitoring

Model drift occurs when the statistical properties of the input data change over time, causing model performance to degrade. In retail forecasting, drift can occur due to seasonal shifts, economic changes, or changes in consumer behavior. Best practices from the MLOps literature recommend:

- Periodic retraining with updated data (e.g., monthly retraining with the latest 2 years of sales history).
- Performance monitoring: tracking rolling MAE on held-out recent data.

- Alerting: automated notifications when performance metrics exceed predefined thresholds.

## 9. Comparative Analysis of Forecasting Methods

Based on the literature review, we summarize the key trade-offs between the forecasting approaches considered for our project:

| Method            | Accuracy    | Scalability | External Regressors | Interpretability | Our Priority  |
|-------------------|-------------|-------------|---------------------|------------------|---------------|
| ARIMA/SARIMA      | Medium      | Low         | Limited (ARIMAX)    | High             | Baseline      |
| XGBoost           | High        | High        | Yes (features)      | Medium           | Primary Model |
| Random Forest     | Medium-High | Medium      | Yes (features)      | Medium-High      | Baseline ML   |
| Linear Regression | Low         | Low         | Yes (features)      | Low              | Baseline ML   |

\* Requires GPU acceleration for efficient training.

### 9.1 Justification of Model Selection

Based on the literature and the characteristics of the Kaggle dataset, our modeling strategy is as follows:

- ARIMA/SARIMA as a univariate statistical baseline: Applied to individual store-product pairs to establish a lower-bound performance benchmark and validate that our dataset exhibits expected temporal patterns.
- XGBoost as the primary model: Extensive feature engineering (lags, rolling stats, promotions, calendar, oil price) transforms the task into a supervised regression problem. XGBoost is expected to be our best-performing model based on competition benchmarks.

## 10. Conclusion

This literature review has surveyed the major forecasting methodologies applicable to our Sales Forecasting and Demand Prediction project. The key conclusions are:

- Classical methods (ARIMA/ SARIMA) provide interpretable baselines and are well-suited for shorter forecast horizons or when data is limited. Prophet's native holiday support makes it particularly relevant for the Ecuadorian retail context.
- Machine learning methods (XGBoost, LightGBM) consistently outperform classical methods on large, multi-store retail datasets when combined with comprehensive feature engineering. They are our primary modeling approach.
- Feature engineering - particularly lag features, rolling statistics, promotion flags, and holiday indicators - is as important as model selection in determining final forecast accuracy.
- MLOps practices (MLflow tracking, Streamlit deployment, Streamlit dashboard, monitoring) are essential for transforming a research model into a production-ready forecasting system.

The literature directly informs our project's technical roadmap: we will implement and compare SARIMA, XGBoost, evaluate them on RMSLE (the Kaggle competition metric), and deploy the best-performing model as an interactive forecasting application.

## 11. References

|     |                                                                                                                                                                                                                                      |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| [8] | Bai, S., Kolter, J.Z., & Koltun, V. (2018). "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling." arXiv:1803.01271.<br>Relevance: TCN vs LSTM comparison.                                 |

|      |                                                                                                                                                                                                                                                                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [9]  | <p>Lim, B., Arik, S.O., Loeff, N., &amp; Pfister, T. (2021). "Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting." <i>International Journal of Forecasting</i>, 37(4), pp. 1748-1764.</p> <p>Relevance: State-of-the-art TFT model.</p> |
| [10] | <p>Fildes, R., Ma, S., &amp; Kolassa, S. (2019). "Retail Forecasting: Research and Practice." <i>International Journal of Forecasting</i>, 38(4), pp. 1283-1318.</p> <p>Relevance: Promotion effects in retail forecasting.</p>                                           |
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| [12] | <p>Zaharia, M., Chen, A., Davidson, A., et al. (2018). "Accelerating the Machine Learning Lifecycle with MLflow." <i>IEEE Data Engineering Bulletin</i>, 41(4), pp. 39-45.</p> <p>Relevance: MLflow for experiment tracking.</p>                                          |

# 03

## Requirements Gathering

Sales Forecasting and Demand Prediction

### 3. Requirements Gathering

#### Requirements Scope

This section describes the requirements for a Streamlit dashboard-based machine learning project. It focuses on stakeholders, user stories, use cases, functional requirements, and non-functional requirements.

#### 1. Requirements Gathering Overview

This section identifies the stakeholders, user needs, user stories, use cases, functional requirements, and non-functional requirements for the Sales Forecasting and Demand Prediction System. The system is designed as a Streamlit dashboard that helps users upload sales data, validate data quality, train forecasting models, generate demand forecasts, analyze trends, and export reports.

The requirements focus on a Data Science workflow and dashboard experience rather than a traditional full-stack application. Therefore, the main system interactions are related to dataset management, data validation, feature engineering, model training, forecast generation, visualization, alerts, and reporting.

#### 2. Stakeholder Analysis

The following stakeholders are involved in the proposed system. Each stakeholder has different goals and information needs.

| Stakeholder               | Main Needs                                                                                                           | System Support                                                                                   |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Data Scientist            | Needs to upload datasets, validate data, prepare features, train models, compare metrics, and monitor model quality. | Dataset Management, Data Validation, Feature Engineering, Model Training, Model Health, Metrics. |
| Inventory Manager         | Needs accurate demand forecasts to avoid stockouts and overstocking.                                                 | Forecast Results, Inventory Risk Score, Stockout Alerts, Weekly Demand Breakdown.                |
| Business Manager          | Needs high-level KPIs, reports, and business insights to support decision-making.                                    | Executive Dashboard, Forecast Accuracy, Predicted Sales, Reports Export, Alerts.                 |
| Marketing Manager         | Needs to understand trends, seasonality, and promotion/holiday impact on demand.                                     | Sales Trends, Holiday Impact Analysis, Demand Spike Alerts, Category Insights.                   |
| Project Team / Developers | Needs a clear structure for implementing the dashboard and machine learning workflow.                                | Modular pipeline design, Streamlit pages, reusable preprocessing and modeling scripts.           |

### 3. User Stories

User stories describe how each stakeholder expects to interact with the dashboard.

| ID    | User Story                                                                                                                                                | Priority |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| US-01 | As a Data Scientist, I want to upload sales datasets so that I can prepare data for forecasting.                                                          | High     |
| US-02 | As a Data Scientist, I want to validate dataset quality so that I can detect missing values, duplicates, outliers, and schema issues before training.     | High     |
| US-03 | As a Data Scientist, I want to generate time, lag, rolling, promotion, holiday, and external-factor features so that the model can learn demand patterns. | High     |
| US-04 | As a Data Scientist, I want to train and compare forecasting models so that I can select the best-performing model.                                       | High     |
| US-05 | As an Inventory Manager, I want to select a store, product/category, and forecast horizon so that I can view future demand.                               | High     |
| US-06 | As an Inventory Manager, I want to see stockout or high-demand alerts so that I can adjust inventory plans.                                               | High     |
| US-07 | As a Business Manager, I want to view an executive dashboard so that I can understand forecast accuracy, predicted sales, risks, and model health.        | Medium   |
| US-08 | As a Business Manager, I want to export forecast reports so that I can share insights with stakeholders.                                                  | Medium   |
| US-09 | As a Marketing Manager, I want to view sales trends and holiday/promotion impact so that I can plan campaigns more effectively.                           | Medium   |
| US-10 | As a Data Scientist, I want to review model metrics such as MAE, RMSE so that I can evaluate forecasting quality.                                         | High     |



## 4. Use Cases

The use cases represent the main interactions between users and the Streamlit dashboard. The use case diagram is included to visualize the system boundary and actor interactions.

Figure 1. Use Case Diagram for the Sales Forecasting and Demand Prediction System.

| Use Case                   | Primary Actor                      | Description                                                                                                                  | Input                                        | Output                                      |
|----------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------|
| Load Sales Dataset         | Data Scientist                     | Loads historical sales and supporting datasets into the dashboard.                                                           | CSV files or local dataset files.            | Loaded dataset.                             |
| Validate Dataset           | Data Scientist                     | Checks dataset quality, required columns, missing values, duplicates, and invalid records.                                   | Raw dataset.                                 | Validation results and detected issues.     |
| Clean & Preprocess Data    | Data Scientist                     | Handles missing values, outliers, duplicates, date conversion, encoding, and scaling if needed.                              | Validated dataset.                           | Cleaned dataset.                            |
| Generate Features          | Data Scientist                     | Creates forecasting features such as date features, lag features, rolling means, promotions, holidays, and external factors. | Cleaned dataset.                             | Feature dataset.                            |
| Train Forecasting Model    | Data Scientist                     | Trains and compares forecasting models such as XGBoost, LightGBM, Random Forest, Prophet, or LSTM.                           | Feature dataset.                             | Trained model and metrics.                  |
| Evaluate Model Performance | Data Scientist                     | Calculates model evaluation metrics and compares model performance.                                                          | Actual values and predictions.               | MAE, RMSE, MAPE/WAPE, and model comparison. |
| Generate Demand Forecast   | Inventory Manager / Data Scientist | Generates future sales and demand forecasts for selected products, stores, or periods.                                       | Product/store/category and forecast horizon. | Forecasted demand values.                   |
| View Forecast Results      | Business / Inventory Manager       | Displays forecasts using charts, tables, projected sales, and anomaly notes.                                                 | Forecast output.                             | Forecast dashboard view.                    |
| View Sales Trends          | Business / Marketing Manager       | Shows historical sales trends, seasonality, and holiday impact analysis.                                                     | Historical sales data.                       | Trend and insight charts.                   |
| Export Forecast Report     | Business / Inventory Manager       | Exports forecast and analysis reports.                                                                                       | Selected report type and date range.         | PDF or CSV report.                          |

## 5. Functional Requirements

Functional requirements describe the features and capabilities that the Streamlit dashboard should provide.

| ID    | Functional Requirement                                                                                                                                           |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-01 | The system shall allow the Data Scientist to upload or load historical sales datasets and related external files.                                                |
| FR-02 | The system shall display available datasets, file type, size, number of rows, last updated date, and processing status.                                          |
| FR-03 | The system shall validate dataset quality, including required columns, date format, missing values, duplicates, outliers, and invalid records.                   |
| FR-04 | The system shall show validation results such as completeness, validity, uniqueness, and detected data issues.                                                   |
| FR-05 | The system shall support data cleaning operations such as handling missing values, removing duplicates, handling outliers, and converting date fields.           |
| FR-06 | The system shall generate forecasting features such as day of week, month, weekend flag, lag features, rolling mean, promotions, holidays, and external factors. |
| FR-07 | The system shall allow the Data Scientist to configure model training settings such as target variable, validation strategy, and optimization metric.            |
| FR-08 | The system shall train and compare forecasting models such as XGBoost, LightGBM, Random Forest, Prophet, and LSTM.                                               |
| FR-09 | The system shall display model evaluation metrics such as WAPE, MAE, RMSE, MAPE, training time, and model health.                                                |
| FR-10 | The system shall allow the user to filter forecast results by store, category, product family, and forecast horizon.                                             |
| FR-11 | The system shall generate demand forecasts and display actual vs forecasted sales charts.                                                                        |
| FR-12 | The system shall display projected sales, forecast accuracy, confidence range, anomaly notes, and weekly forecast breakdowns.                                    |
| FR-13 | The system shall display sales trends, monthly comparisons, top regions, best categories, and holiday impact analysis.                                           |
| FR-14 | The system shall display alerts for high stockout risk, unexpected demand spikes, model performance drops, pipeline delays, and retraining completion.           |
| FR-15 | The system shall allow users to export reports in PDF or CSV format.                                                                                             |
| FR-16 | The system shall allow users to configure dashboard settings such as profile information, role/default view, and notification preferences.                       |

## 6. Non-Functional Requirements

Non-functional requirements define the quality criteria of the dashboard and forecasting workflow.

| Category         | Requirement                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usability        | The Streamlit dashboard should be clear, simple, and easy to navigate using a sidebar layout and readable dashboard pages.                                                                   |
| Performance      | The dashboard should load pages and display charts within an acceptable prototype response time. Heavy model training may take longer and can be handled as a separate run.                  |
| Accuracy         | Forecasting models should be evaluated using standard metrics such as MAE, RMSE, MAPE, or WAPE. The selected model should be based on comparative performance.                               |
| Reliability      | The system should handle invalid datasets, missing values, and failed training runs without crashing and should display clear error messages.                                                |
| Maintainability  | The project code should be organized into clear modules or scripts for data loading, validation, preprocessing, feature engineering, training, evaluation, forecasting, and dashboard pages. |
| Scalability      | The design should allow adding more stores, product categories, features, external factors, and forecasting models in future phases.                                                         |
| Security         | Since the current version is a prototype Streamlit dashboard, advanced authentication is not included. If deployed later, access control and secure file handling should be added.           |
| Reproducibility  | Model runs, metrics, parameters, and data versions should be documented or tracked where possible using notebooks, GitHub, and optional MLflow.                                              |
| Availability     | The prototype should be runnable locally or on a simple Streamlit deployment environment during demonstration.                                                                               |
| Interpretability | The dashboard should present forecasts, model metrics, alerts, and trends in a way that business users can understand.                                                                       |

## 7. Assumptions and Constraints

| Item                | Description                                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation Type | The system is a Streamlit dashboard prototype supported by a machine learning pipeline.                                                         |
| Data Storage        | No physical database is implemented in the current phase. Data is loaded from CSV files and processed using dataframes.                         |
| API                 | No REST API or backend endpoint layer is included. Therefore, API documentation and API testing are not applicable.                             |
| Testing Scope       | Testing focuses on data validation, preprocessing correctness, model evaluation, forecast output review, and manual dashboard usability checks. |
| OOP Design          | The project does not require a class diagram because the implementation is not based on OOP classes.                                            |
| State Diagram       | A state diagram is not required because the project workflow is better represented using activity diagrams, sequence diagrams, and DFDs.        |

## 8. Acceptance Criteria

| Requirement Area    | Acceptance Criteria                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Dataset Management  | A user can load the required sales dataset files and see their status in the dashboard.                                          |
| Data Validation     | The dashboard displays validation results and detected issues such as missing values, duplicates, outliers, and schema problems. |
| Feature Engineering | The system creates time-based, lag, rolling, promotion, holiday, and external factor features.                                   |
| Model Training      | At least one forecasting model is trained and evaluated, and model metrics are displayed.                                        |
| Forecast Results    | A user can select a forecast horizon and view predicted demand using charts and tables.                                          |
| Sales Trends        | The dashboard displays historical sales trends and relevant business insights.                                                   |
| Alerts              | The dashboard displays meaningful alerts such as stockout risk, demand spikes, or model performance drops.                       |
| Reports             | A user can export forecast-related results or summaries as PDF or CSV.                                                           |

## 9. Requirements Gathering Summary

The Requirements Gathering section defines the users, needs, user stories, use cases, functional requirements, and non-functional requirements of the Sales Forecasting and Demand Prediction System. The requirements are aligned with the current scope of the project as a Streamlit dashboard and machine learning forecasting pipeline. The design intentionally excludes a physical database, backend API, API testing, OOP class diagram, and state diagram because these are not part of the current implementation scope.

## 10. Source Files Used

- IBM Data Science Project - Round4.pdf: Project 5 requirements for Sales Forecasting and Demand Prediction.
- Project Documentation.pdf: Requirements Gathering documentation requirements.
- Uploaded use case diagram and dashboard UI/UX mockups prepared for the project.
- Architecture Style: Machine Learning Pipeline Architecture document and software architecture diagram.

# 04

## System Analysis & Design

Sales Forecasting and Demand Prediction

## 4. System Analysis & Design

### Design Scope

This section excludes state and class diagrams because the current project is not implemented using complex object states or OOP class design. It also excludes database design and API testing because the implementation is a Streamlit dashboard using CSV/dataframes rather than a backend service.

- 1. Introduction
- 2. Problem Statement
- 3. System Objectives
- 4. System Scope
- 5. System Boundary
- 6. Stakeholders and Actors
- 7. Requirements
- 8. System Architecture
- 9. Use Case Diagram
- 10. Activity Diagram
- 11. Sequence Diagrams
- 12. Data Flow Diagrams
- 13. UI/UX Dashboard Design
- 14. Data Storage, API, Testing, State and Class Diagram Notes
- 15. Technology Stack
- 16. Summary

## Design Scope

This section excludes state and class diagrams because the current project is not implemented using complex object states or OOP class design. It also excludes database design and API testing because the implementation is a Streamlit dashboard using CSV/dataframes rather than a backend service.

## 1. Introduction

The Sales Forecasting and Demand Prediction System is a proposed analytical dashboard and machine learning pipeline designed to predict future product demand using historical sales data and external factors. The system supports business decision-making by converting raw sales datasets into forecasts, model metrics, alerts, and reports.

This project is not designed as a traditional full-stack transactional application. Instead, it is a Data Science project whose practical interface is a Streamlit dashboard. Therefore, the System Analysis and Design section focuses on the forecasting workflow, dashboard interactions, data flow, machine learning pipeline, and reporting outputs.

## 2. Problem Statement

Retail and e-commerce businesses often face difficulty in accurately predicting future sales and demand. Inaccurate demand forecasting may cause stockouts, overstocking, weak inventory planning, inefficient marketing decisions, and poor operational planning. The proposed dashboard-based forecasting system addresses this problem by using historical sales patterns and external factors to generate future demand forecasts.

## 3. System Objectives

Analyze historical sales data and identify sales trends, seasonality, and demand patterns.

Validate, clean, and prepare datasets before using them in forecasting models.

Generate useful features such as time features, lag features, rolling means, promotion flags, holiday indicators, and external factors.

Train and compare forecasting models such as XGBoost, LightGBM, Random Forest, Prophet, and LSTM.

Evaluate model performance using metrics such as MAE, RMSE, MAPE, or WAPE.

Generate future demand forecasts for selected stores, products, categories, and forecast periods.

Display results through a Streamlit dashboard with KPIs, charts, alerts, and reports.

Support business users in inventory planning, marketing decisions, and operational forecasting.

## 4. System Scope

### 4.1 In Scope

- Loading sales datasets from CSV files and related structured files.
- Validating dataset format, required columns, missing values, duplicates, outliers, and invalid records.
- Preprocessing and cleaning the data.
- Creating forecasting features and merging external factors such as holidays, oil prices, promotions, stores, and transactions.
- Training and comparing machine learning and forecasting models.
- Generating demand forecasts and displaying them through a Streamlit dashboard.
- Exporting analytical reports as PDF or CSV files.



- Displaying dashboard pages for dataset management, validation, feature engineering, training, forecasts, trends, alerts, reports, and settings.

#### 4.2 Out of Scope for the Current Phase

- A physical production database design or implemented DBMS.
- A REST API or backend service layer.
- API testing, endpoint testing, or backend integration testing.
- Object-oriented implementation based on classes.
- State diagram and class diagram deliverables, because the implementation is not OOP-based and does not manage complex object state transitions.
- Full cloud deployment, enterprise authentication, ERP/POS integration, and production monitoring.

### 5. System Boundary

| Category   | Description                                                                                                                                    |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Inputs     | Historical sales data, store data, product/family data, promotions, holidays, transactions, oil prices, and user forecast requests.            |
| Processing | Validation, cleaning, preprocessing, feature engineering, model training, model evaluation, forecast generation, visualization, and reporting. |
| Outputs    | Demand forecasts, forecast charts, actual vs predicted views, model metrics, alerts, and exported reports.                                     |

### 6. Stakeholders and Actors

| Stakeholder / Actor | Role in the System                                                                                                                         |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Data Scientist      | Loads datasets, validates data, performs preprocessing, engineers features, trains models, compares performance, and reviews model health. |
| Inventory Manager   | Uses forecasts to plan inventory levels, detect stockout risks, and support restocking decisions.                                          |
| Business Manager    | Uses dashboard KPIs, forecasts, alerts, and reports to support business decisions.                                                         |
| Marketing Manager   | Uses sales trends, seasonality, promotion impact, and demand patterns to improve marketing planning.                                       |

## 7. Requirements

### 7.1 Functional Requirements

| ID    | Functional Requirement                                                                                                                     |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------|
| FR-01 | The system shall allow the Data Scientist to upload or load historical sales datasets.                                                     |
| FR-02 | The system shall validate dataset structure, required columns, and data quality.                                                           |
| FR-03 | The system shall detect missing values, duplicates, outliers, and invalid records.                                                         |
| FR-04 | The system shall clean and preprocess sales data before modeling.                                                                          |
| FR-05 | The system shall merge sales data with external factors such as holidays, oil prices, promotions, stores, and transactions when available. |
| FR-06 | The system shall generate time-based, lag, rolling, promotion, holiday, and external factor features.                                      |
| FR-07 | The system shall train forecasting models and compare their performance.                                                                   |
| FR-08 | The system shall allow users to select product, store, category, and forecast period.                                                      |
| FR-09 | The system shall generate demand forecasts and display forecast results.                                                                   |
| FR-10 | The system shall display model metrics such as MAE, RMSE>                                                                                  |
| FR-11 | The system shall display alerts such as stockout risk, demand spikes, model performance drops, and data pipeline issues.                   |
| FR-12 | The system shall export forecast reports in PDF or CSV format.                                                                             |

## 7.2 Non-Functional Requirements

| Requirement Type | Description                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Usability        | The Streamlit dashboard should be simple, readable, and easy to navigate.                                                                        |
| Accuracy         | Forecasting performance should be evaluated using accepted forecasting metrics.                                                                  |
| Performance      | Dashboard views and forecast calculations should respond within acceptable prototype time.                                                       |
| Reliability      | The system should handle invalid data and failed model runs without crashing.                                                                    |
| Maintainability  | The pipeline should be modular and separated into data preparation, feature engineering, training, evaluation, forecasting, and reporting steps. |
| Scalability      | The design should support additional products, stores, models, and external factors in future work.                                              |
| Reproducibility  | Experiments, parameters, metrics, and datasets should be trackable where possible using tools such as MLflow.                                    |
| Interpretability | The dashboard should present forecasts, metrics, and alerts in a clear business-friendly format.                                                 |

## 8. System Architecture

### 8.1 Architecture Style

The proposed system follows a Machine Learning Pipeline Architecture. The pipeline processes data sequentially: raw data is collected, cleaned, transformed, used for feature engineering, passed to model training and evaluation, then used to generate predictions and dashboard visualizations.

The uploaded architecture description defines the main components as data sources, data processing, feature engineering, model training, evaluation, prediction, deployment/dashboard layer, and monitoring/retraining. The source datasets include train, test, sample submission, stores, oil, holidays, and transactions files. The architecture also includes cleaning, merging, date conversion, encoding, scaling, lag/rolling features, XGBoost, Linear Regression, Random Forest, MLflow tracking, and future monitoring/retraining.

## 8.2 Architecture Layers

| Layer                         | Description                                                                                                                           |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Data Sources Layer            | CSV files such as train.csv, test.csv, stores.csv, sample_submission.csv, oil.csv, holidays, and transactions.                        |
| Preprocessing Layer           | Handles missing values, outliers, duplicate removal, dataset merging, date conversion, encoding, and normalization/scaling.           |
| Feature Engineering Layer     | Creates time features, lag features, rolling means, promotion indicators, holiday features, oil price features, and external factors. |
| Model Training Layer          | Trains forecasting models such as XGBoost, Random Forest, SARIMA, and Linear Regression.                                              |
| Evaluation Layer              | Compares models using RMSE, MAE, and other forecasting metrics.                                                                       |
| Prediction Layer              | Generates future sales and demand predictions.                                                                                        |
| Dashboard Layer               | Displays predictions, visualizations, alerts, and reports through a Streamlit dashboard.                                              |
| Monitoring & Retraining Layer | Future extension for model performance monitoring, data drift detection, periodic retraining, and updated evaluation.                 |

## 9. Use Case Diagram

The use case diagram shows the interactions between the main actors and the forecasting dashboard system. It focuses on data science workflow actions and business usage actions.

| Use Case                   | Actor                              | Description                                                            |
|----------------------------|------------------------------------|------------------------------------------------------------------------|
| Load Sales Dataset         | Data Scientist                     | Loads the historical sales dataset into the system.                    |
| Validate Dataset           | Data Scientist                     | Checks structure, required fields, and quality problems.               |
| Clean & Preprocess Data    | Data Scientist                     | Handles missing values, duplicates, outliers, and data formatting.     |
| Generate Features          | Data Scientist                     | Creates date, lag, rolling, promotion, holiday, and external features. |
| Train Forecasting Model    | Data Scientist                     | Trains machine learning forecasting models.                            |
| Evaluate Model Performance | Data Scientist                     | Reviews model metrics and compares performance.                        |
| Generate Demand Forecast   | Inventory Manager / Data Scientist | Generates future demand predictions.                                   |
| View Forecast Results      | Business / Inventory Manager       | Views predicted demand charts and tables.                              |
| View Sales Trends          | Business / Marketing Manager       | Explores historical trends and seasonal patterns.                      |
| Export Forecast Report     | Business / Inventory Manager       | Exports forecast summaries and analytical reports.                     |

## 10. Activity Diagram

The swimlane activity diagram divides the process between the Data Scientist, the System/ML Pipeline, and the Business or Inventory Manager. The workflow starts with loading a historical dataset, validating it, cleaning and preprocessing it, generating features, training and evaluating models, selecting the best model, generating future demand forecasts, visualizing results, and exporting the forecast report.

## 11. Sequence Diagrams

### 11.1 Generate Demand Forecast

This sequence describes the main forecasting use case. The Inventory Manager selects a product and forecast period. The forecasting interface sends the request to the ML pipeline. The system validates data, preprocesses it, generates features, predicts future demand, prepares visualizations, and displays the demand forecast. If the dataset is invalid, validation errors are returned.

## 11.2 Export Forecast Report

This sequence shows the report export process. The Business Manager requests a forecast report. The system checks whether forecast results are available, retrieves forecast values and charts, generates a PDF or CSV report, and returns it to the user. If no forecast exists, the user is asked to generate a forecast first.

## 11.3 Train Forecasting Model

This sequence shows the model training use case. The Data Scientist starts a training run, the ML pipeline prepares the dataset, the forecasting model is trained, and the evaluation module calculates performance metrics. If the performance is not acceptable, the model is tuned and retrained.

# 12. Data Flow Diagrams

## 12.1 Context-Level DFD

The context-level DFD shows the Sales Forecasting and Demand Prediction System as a single process. The Data Scientist provides uploaded sales data, the data source provides historical records, and the Business/Inventory Manager sends forecast requests. The system returns validation results, metrics, demand forecasts, and reports.

## 12.2 Level-1 DFD

The Level-1 DFD decomposes the system into five main processes: Data Preparation, Feature Engineering, Model Training, Forecast Generation, and Reporting & Visualization. It shows how the sales dataset moves through the pipeline until forecast results, charts, reports, and model metrics are produced.

### 13. UI/UX Dashboard Design

The proposed UI/UX is a Streamlit dashboard rather than a full application. The dashboard is designed to support both technical users and business users. It contains pages for dataset management, validation, feature engineering, model training, forecast results, trends, alerts, report generation, and settings.

| Dashboard Page           | Primary User                  | Purpose                                                                                                                             |
|--------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Executive Dashboard      | Business / Inventory Manager  | Shows forecast accuracy, predicted sales, inventory risk, active models, demand chart, store performance, alerts, and model health. |
| Dataset Management       | Data Scientist                | Uploads and manages dataset files and data source status.                                                                           |
| Data Validation          | Data Scientist                | Displays completeness, validity, uniqueness, missing values, outliers, and validation rules.                                        |
| Feature Engineering      | Data Scientist                | Shows time features, lag features, rolling windows, external holiday features, and feature summary.                                 |
| Model Training Workspace | Data Scientist                | Configures training runs and compares forecasting algorithms.                                                                       |
| Forecast Results         | Business / Inventory Manager  | Shows demand forecasts, projected sales, accuracy, anomaly notes, and weekly forecast breakdown.                                    |
| Sales Trends & Insights  | Business / Marketing Manager  | Shows historical trends, monthly comparisons, seasonal patterns, and holiday impact analysis.                                       |
| Alerts & Exceptions      | Business / Inventory Manager  | Shows stockout risk, demand spikes, model performance drops, pipeline delays, and retraining completion.                            |
| Report Generation        | Business Manager              | Exports executive, operational, model performance, and exception reports.                                                           |
| Platform Settings        | Data Scientist / System Owner | Manages profile, default role/view, notification preferences, and system options.                                                   |

### 14. Data Storage, API, Testing, State and Class Diagram Notes

#### 14.1 Database Note

A physical database is not included in the current system design. The prototype is a Streamlit dashboard that works with CSV files and in-memory dataframes. Therefore, a full ERD, physical schema, normalization design, and database implementation are not part of the current phase. If the project is extended later, a database can be added to store products, stores, sales records, model runs, and forecast results.

## 14.2 API and Testing Note

No REST API or backend service is included in the current phase because the proposed implementation is a Streamlit dashboard. Therefore, API endpoint documentation and API testing are not applicable. Testing is limited to dashboard-level and data-science validation, such as dataset validation checks, preprocessing checks, model metric validation, forecast output review, and manual dashboard usability testing.

## 14.3 State Diagram Note

A state diagram is not included because the current Streamlit prototype does not manage complex object states or transactional state transitions. The system is pipeline-oriented and its behavior is better represented using activity diagrams, sequence diagrams, and data flow diagrams.

## 14.4 Class Diagram Note

A class diagram is not included because the current implementation is not based on an object-oriented design. The system is structured as a data science pipeline and dashboard workflow rather than OOP classes. The main design components are represented through the software architecture diagram and DFDs instead.

## 15. Technology Stack

| Layer           | Suggested Technology                            | Purpose                                                    |
|-----------------|-------------------------------------------------|------------------------------------------------------------|
| Programming     | Python                                          | Main language for data analysis and machine learning.      |
| Data Handling   | Pandas, NumPy                                   | Loading, cleaning, transforming, and analyzing datasets.   |
| Models          | XGBoost, LightGBM, Random Forest, Prophet, LSTM | Forecasting model training and comparison.                 |
| Evaluation      | Scikit-learn metrics                            | MAE, RMSE, MAPE/WAPE, and model comparison.                |
| Visualization   | Plotly, Matplotlib, Streamlit charts            | Dashboard charts and analytical visualizations.            |
| Dashboard       | Streamlit                                       | Prototype dashboard for user interaction and reporting.    |
| Tracking        | MLflow (optional)                               | Model experiment tracking and versioning.                  |
| Version Control | GitHub                                          | Project documentation, notebooks, code, and collaboration. |



## 16. Summary

The Sales Forecasting and Demand Prediction System is designed as a Streamlit dashboard supported by a machine learning pipeline. The design focuses on data preparation, feature engineering, model training, evaluation, forecast generation, visualization, alerts, and report export. Since the current scope does not include a database, API, backend service, or OOP implementation, the documentation excludes physical database design, API testing, state diagram, and class diagram. The provided diagrams and UI/UX mockups sufficiently represent the system behavior, user interactions, data flow, forecasting workflow, and dashboard interface.

## 17. Source Files Used

- IBM Data Science Project - Round4.pdf: Project 5 requirements for Sales Forecasting and Demand Prediction.
- Project Documentation.pdf: System Analysis & Design documentation guidelines.
- Architecture Style\_ Machine Learning Pipeline Architecture.docx: Architecture description and component list.
- Software Architecture.drawio (2).pdf: Software architecture diagram.
- Uploaded screenshots: Use case, activity, sequence, DFD diagrams, and UI/UX dashboard mockups.

## Appendix A. Diagrams and UI/UX Screens

This appendix includes the diagrams and dashboard mockups used to support the Requirements Gathering and System Analysis & Design sections.

**Figure A1. Software Architecture: Machine Learning Pipeline Architecture**

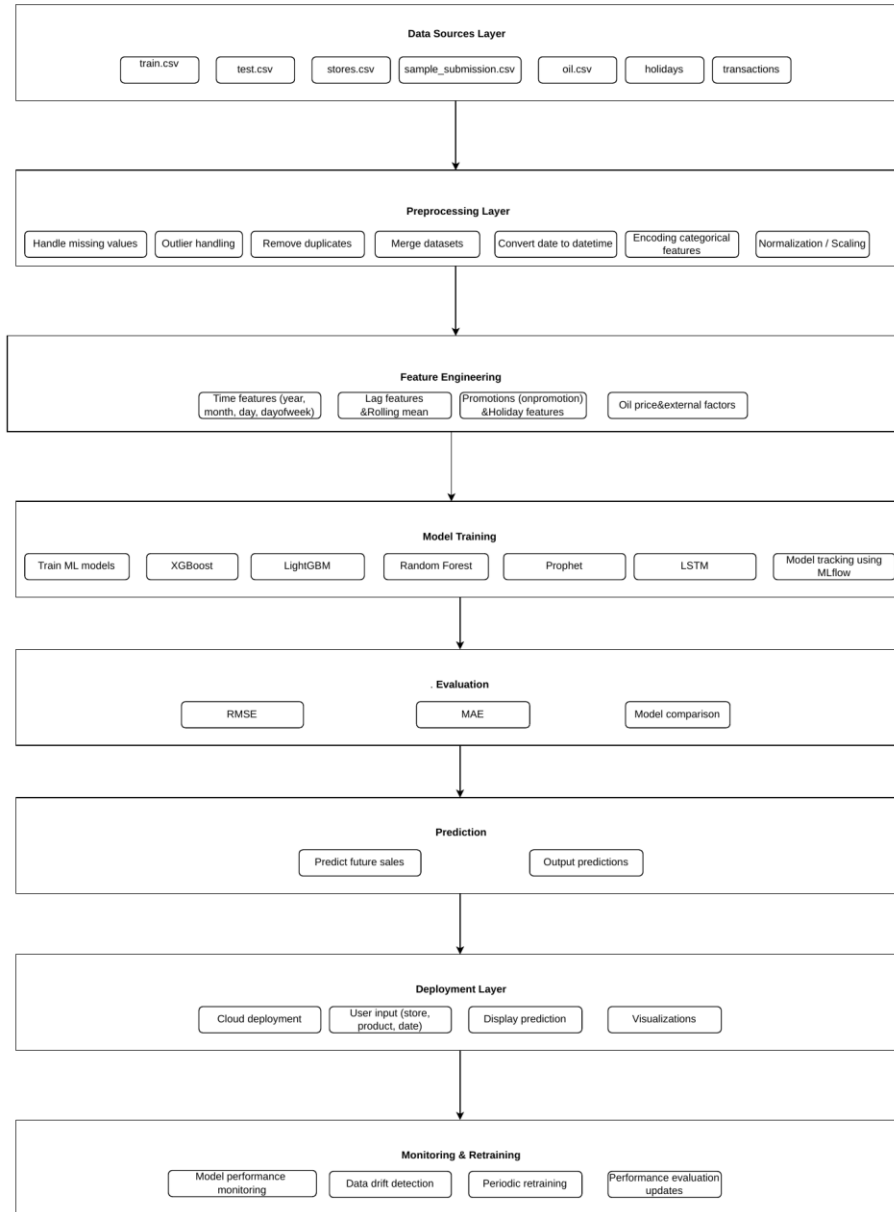
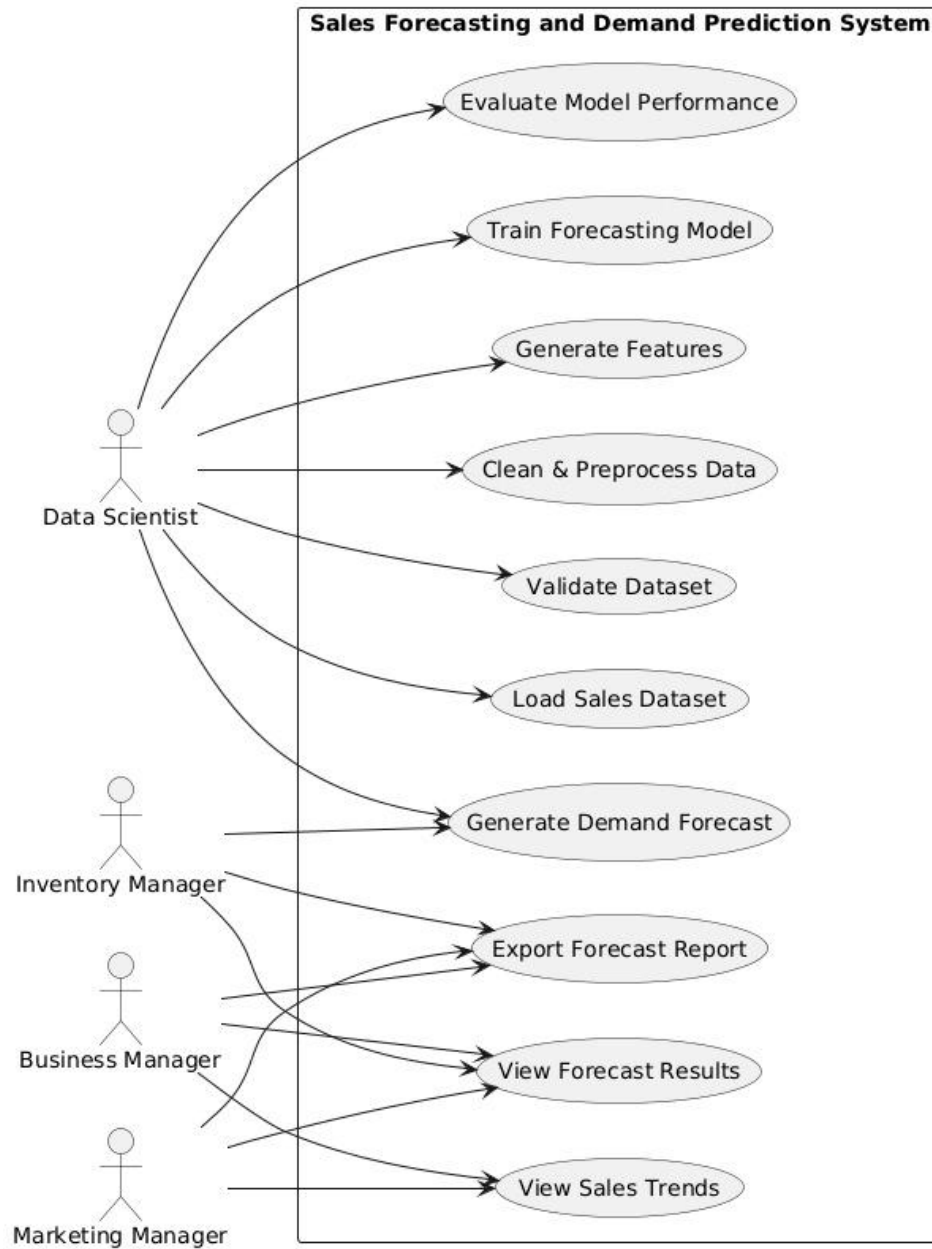
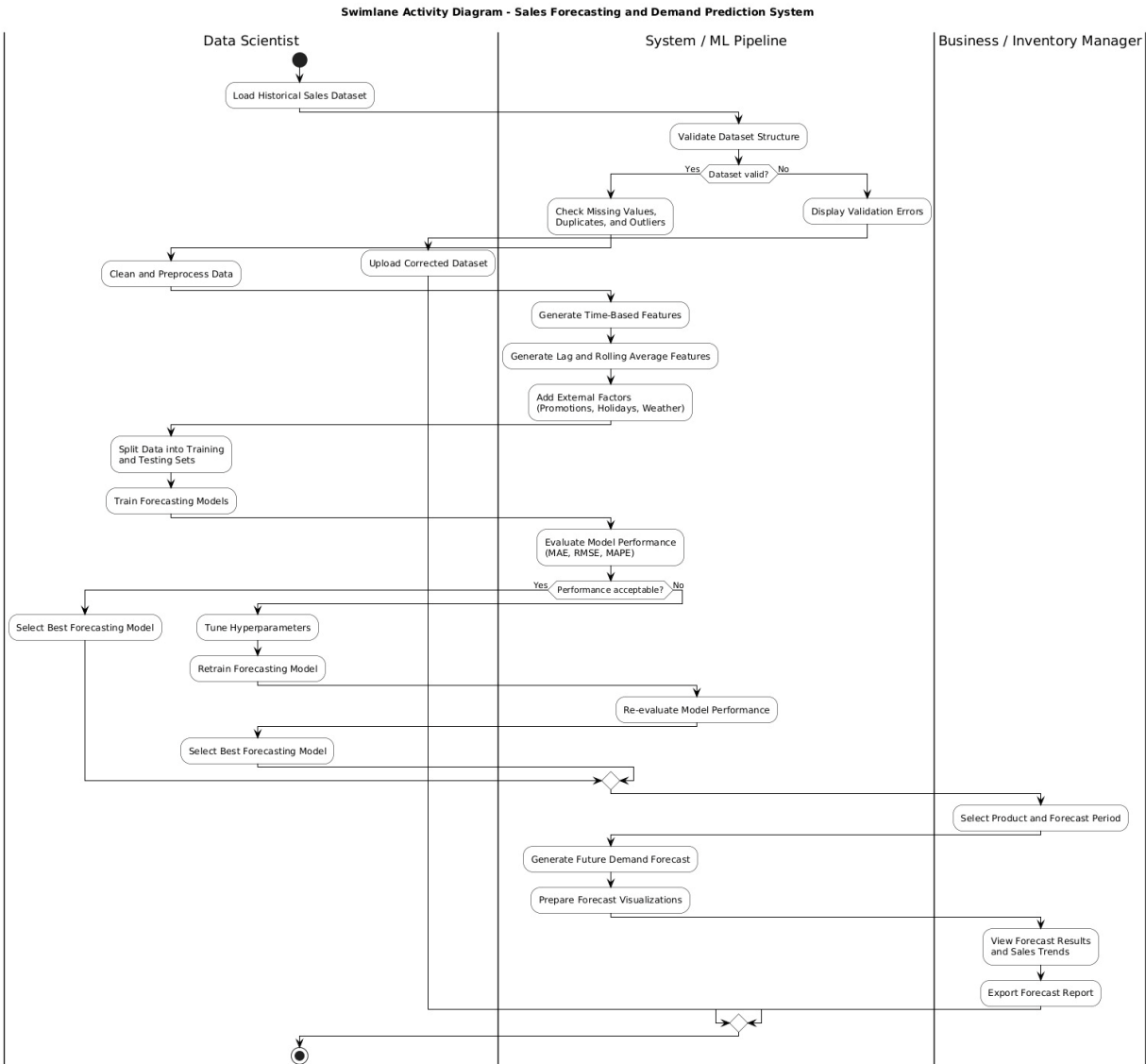
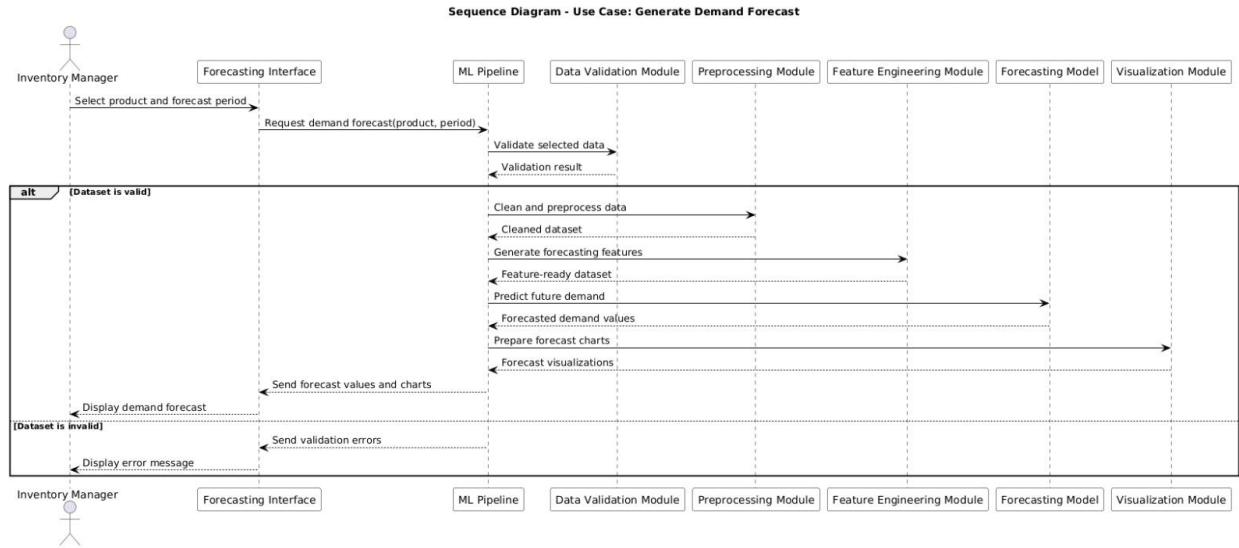


Figure A2. Use Case Diagram

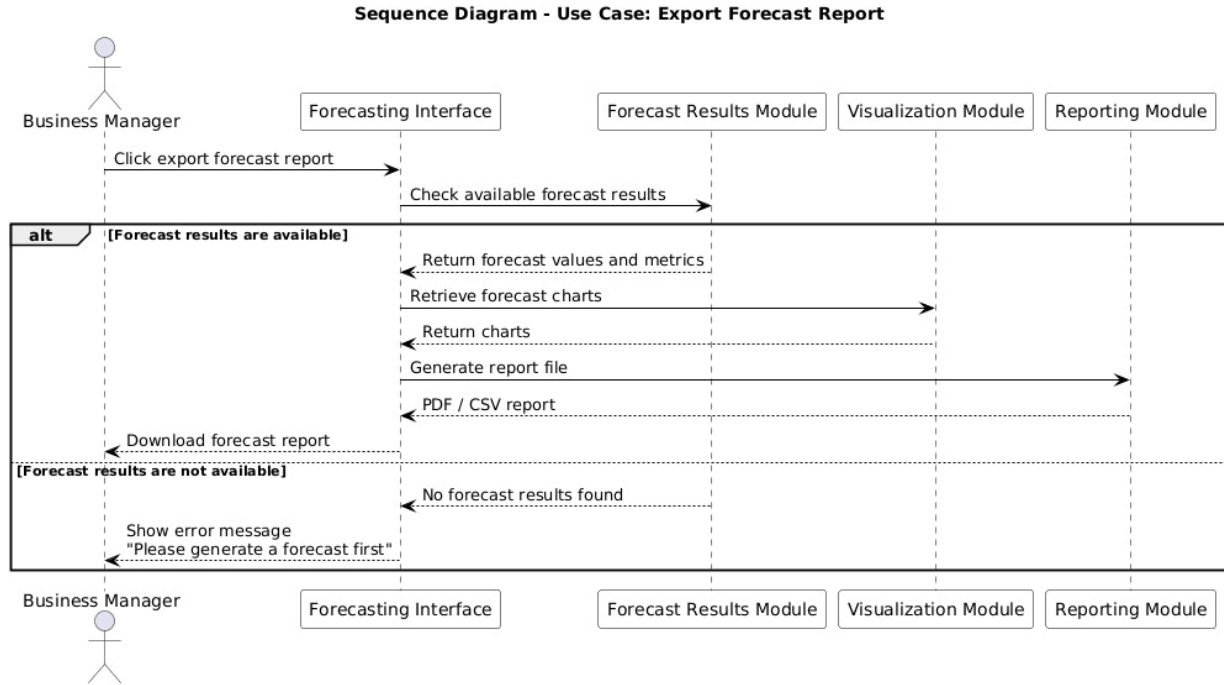


**Figure A3. Swimlane Activity Diagram**



**Figure A4. Sequence Diagram - Generate Demand Forecast**

**Figure A5. Sequence Diagram - Export Forecast Report**



**Figure A6. Sequence Diagram - Train Forecasting Model**

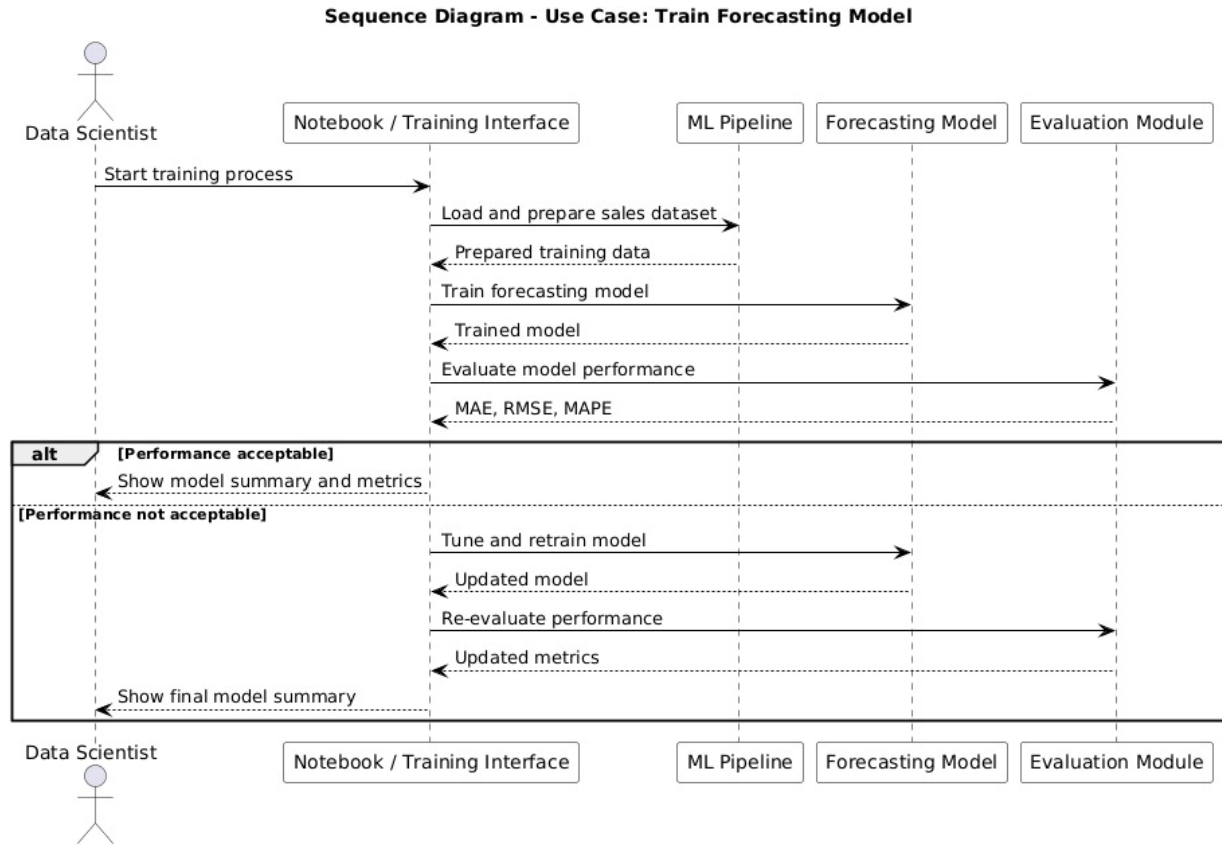
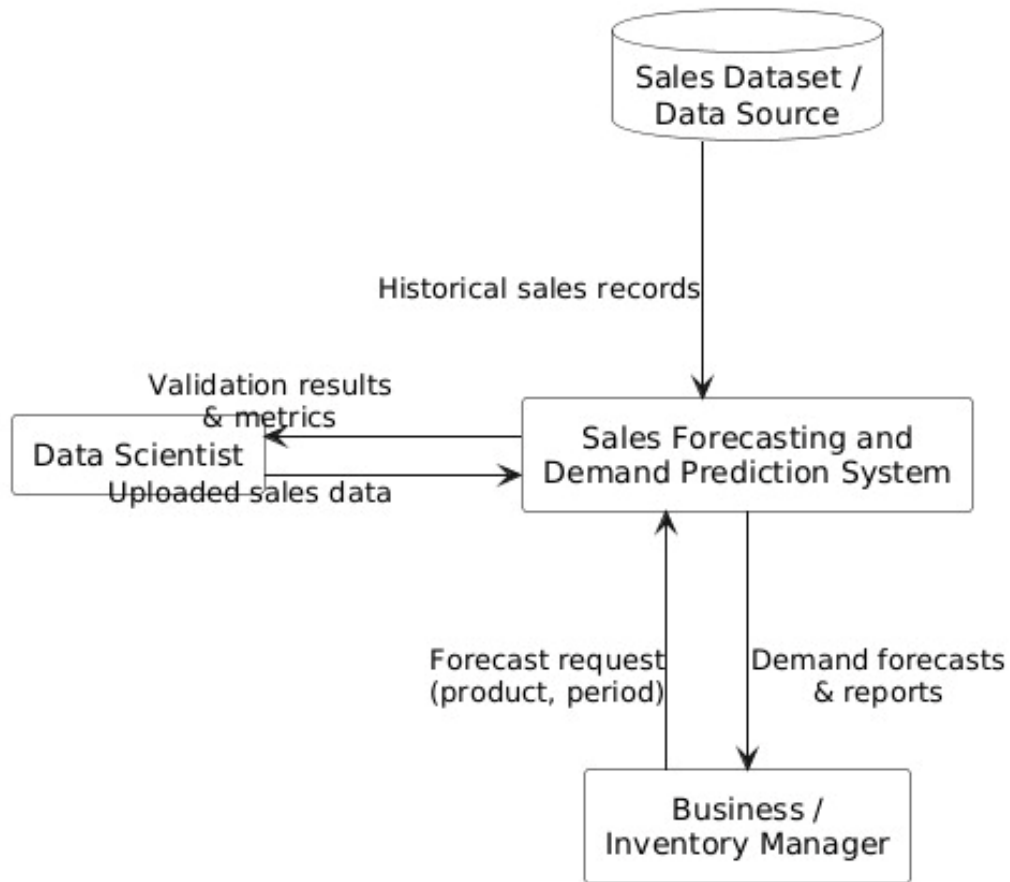


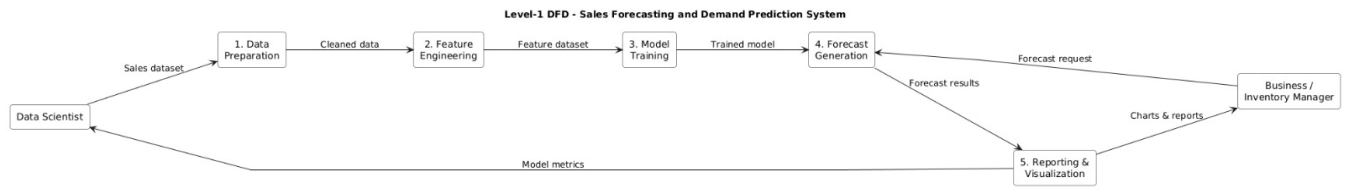
Figure A7. Context-Level DFD

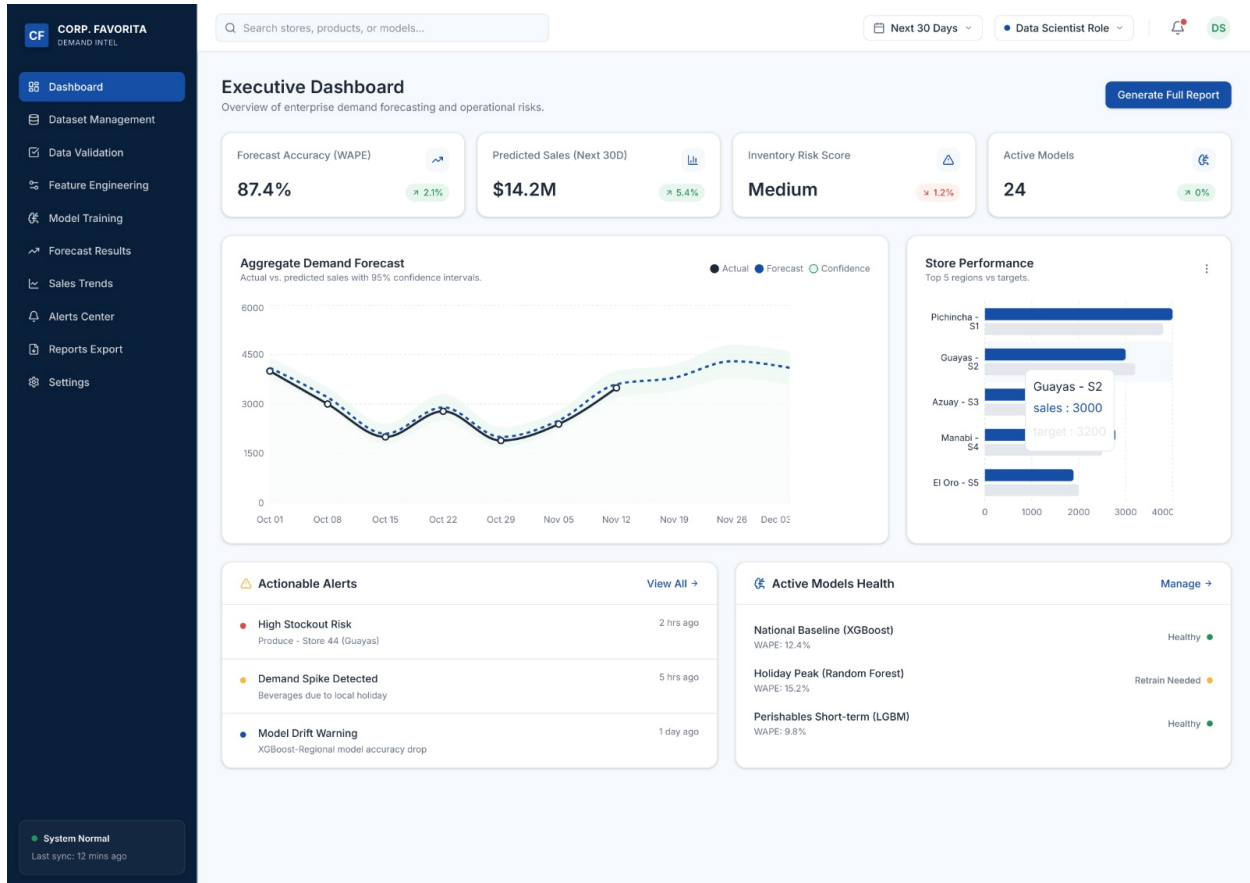
### Context-Level DFD - Sales Forecasting and Demand Prediction System





**Figure A8. Level-1 DFD**



**Figure A9. Executive Dashboard Mockup**

**Figure A10. Dataset Management Mockup**

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Dashboard

**Dataset Management**

Data Validation

Feature Engineering

Model Training

Forecast Results

Sales Trends

Alerts Center

Reports Export

Settings

System Normal

Last sync: 12 mins ago

Q Search stores, products, or models...

Next 30 Days

Data Scientist Role

DS

### Dataset Management

Ingest, monitor, and manage data sources for model training.

Click to upload or drag and drop

CSV, Parquet, or Excel files (Max 5GB)

Select Files

Data Warehouse  
Connected (Snowflake)

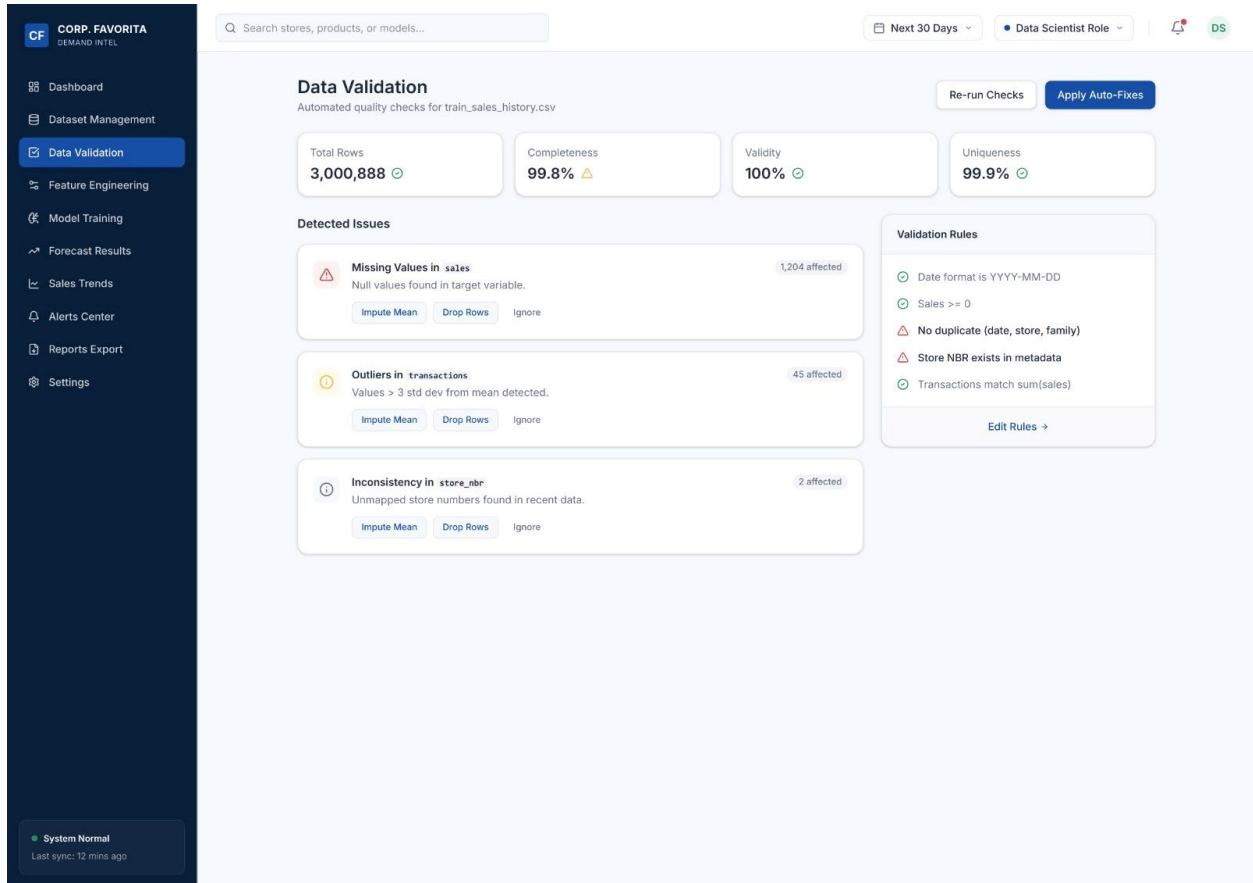
Auto-sync Pipeline  
Active (Daily at 02:00)

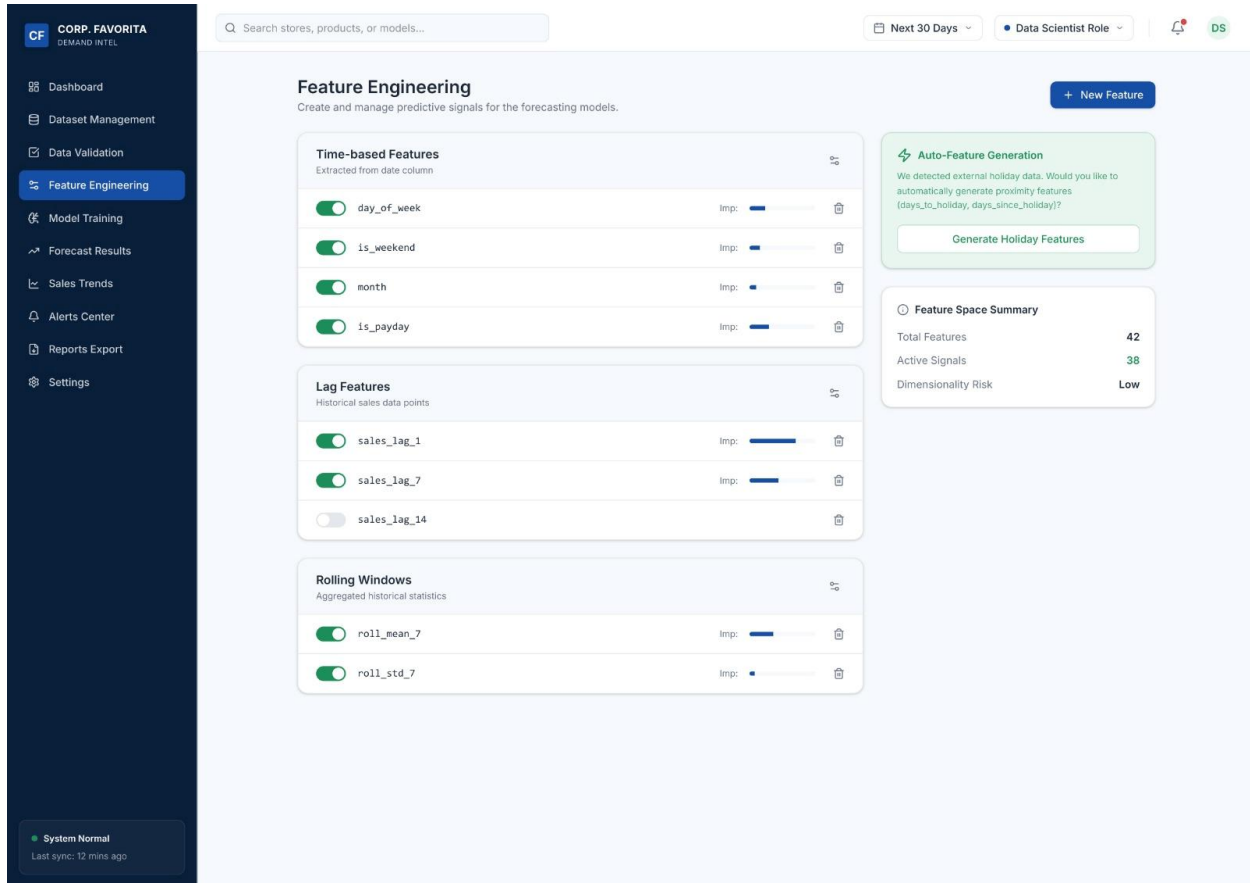
Schema Validation  
1 Warning Detected

Available Datasets

Filter files...

| FILENAME                 | TYPE         | SIZE   | ROWS | LAST UPDATED    | STATUS     |
|--------------------------|--------------|--------|------|-----------------|------------|
| train_sales_history.csv  | Transactions | 2.4 GB | 3M+  | Today, 04:00 AM | Ready      |
| stores_metadata.csv      | Dimension    | 12 KB  | 54   | Oct 1, 2023     | Ready      |
| daily_oil_prices.csv     | External     | 45 KB  | 1.2K | Today, 06:15 AM | Ready      |
| national_holidays_ec.csv | External     | 18 KB  | 350  | Jan 1, 2023     | Ready      |
| promotions_q4.csv        | Campaign     | 4.1 MB | 125K | Just now        | Processing |

**Figure A11. Data Validation Mockup**

**Figure A12. Feature Engineering Mockup**

**Figure A13. Model Training Workspace Mockup**

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Dashboard

Dataset Management

Data Validation

Feature Engineering

**Model Training**

Forecast Results

Sales Trends

Alerts Center

Reports Export

Settings

System Normal

Last sync: 12 mins ago

Q Search stores, products, or models...

Next 30 Days

Data Scientist Role

DS

Model Training Workspace

Configure, train, and compare forecasting algorithms.

Start New Run

Run Configuration

Target Variable

sales

Validation Strategy

Time Series Split (3 folds)

Optimization Metric

WAPE (Weighted MAPE)

Hyperparameters

Auto-Tune On

Bayesian optimization will search the parameter space for 50 trials.

Experiment Leaderboard

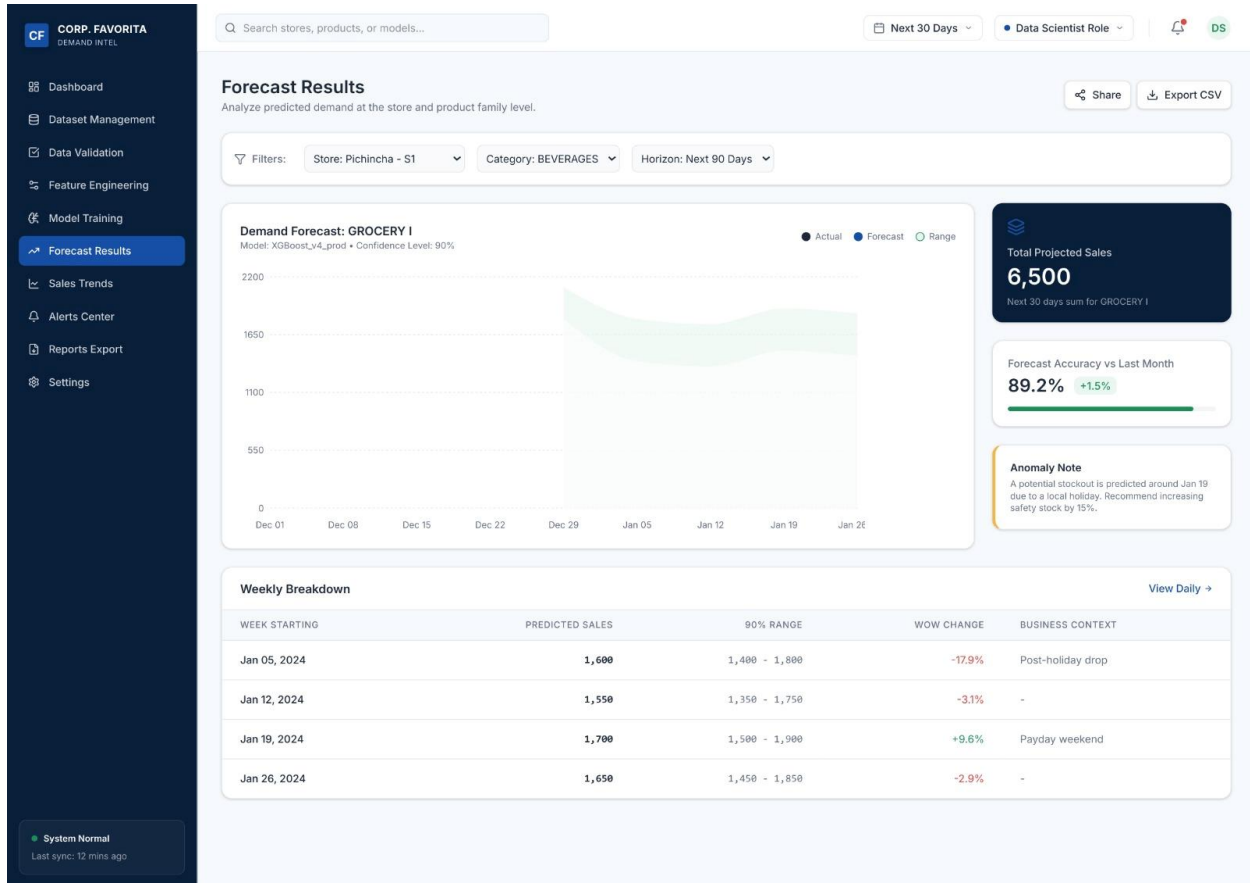
Run ID: #EXP-089

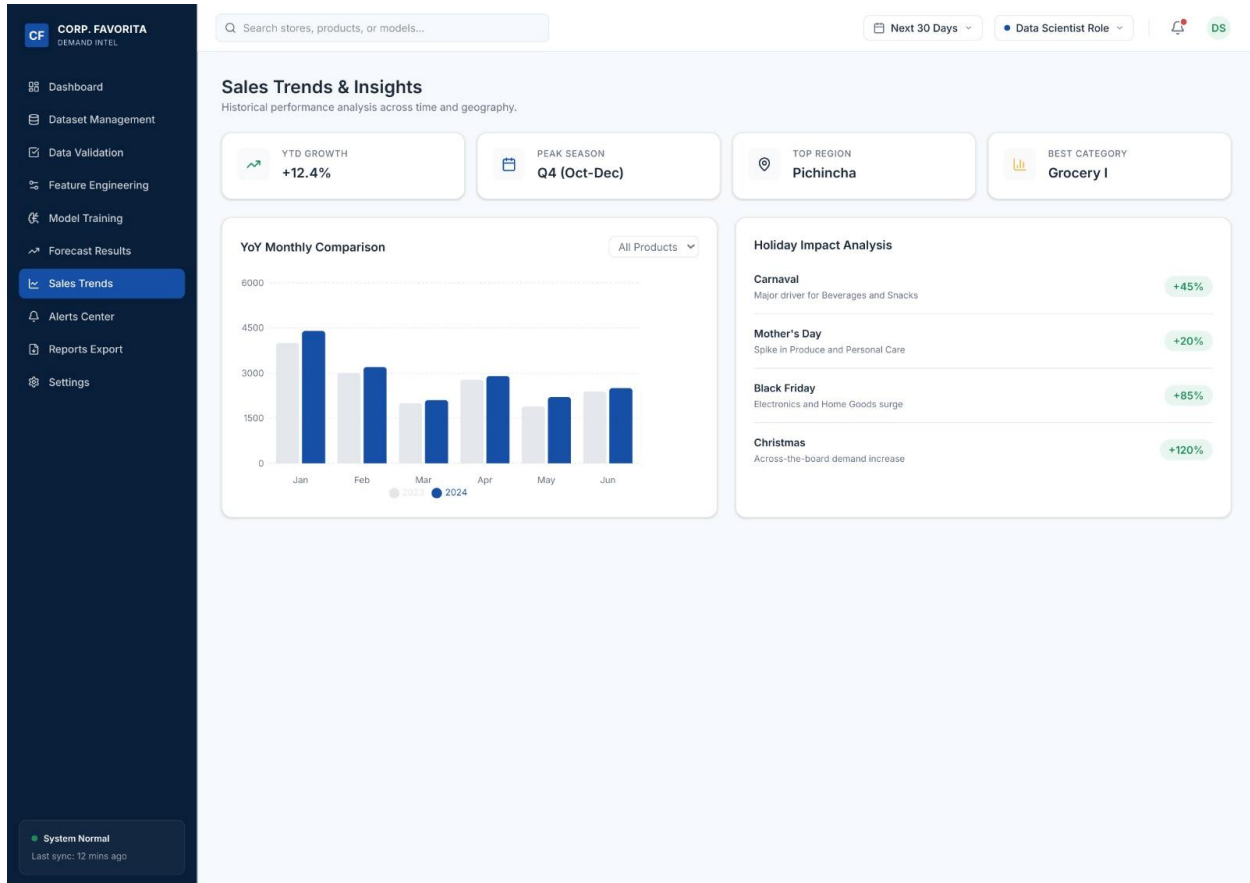
| MODEL NAME                                | WAPE  | RMSE  | TRAIN TIME | ACTION              |
|-------------------------------------------|-------|-------|------------|---------------------|
| <b>XGBoost Regressor</b><br>Tree Ensemble | 12.4% | 420.5 | 45m        | <b>Deploy Model</b> |
| LightGBM<br>Tree Ensemble                 | 13.1% | 445.2 | 18m        | Compare             |
| Prophet<br>Time Series                    | 18.5% | 588.1 | 1h 20m     | Compare             |
| Linear Regression<br>Baseline             | 24.2% | 898.4 | 2m         | Compare             |

AutoML Recommendation

**XGBoost Regressor** achieved the lowest WAPE. It handles the non-linear holiday effects better than the baseline models. Ready for production deployment.

Figure A14. Forecast Results Mockup



**Figure A15. Sales Trends & Insights Mockup**



**Figure A16. Alerts & Exceptions Mockup**

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Search stores, products, or models...

Next 30 Days Data Scientist Role DS

### Alerts & Exceptions

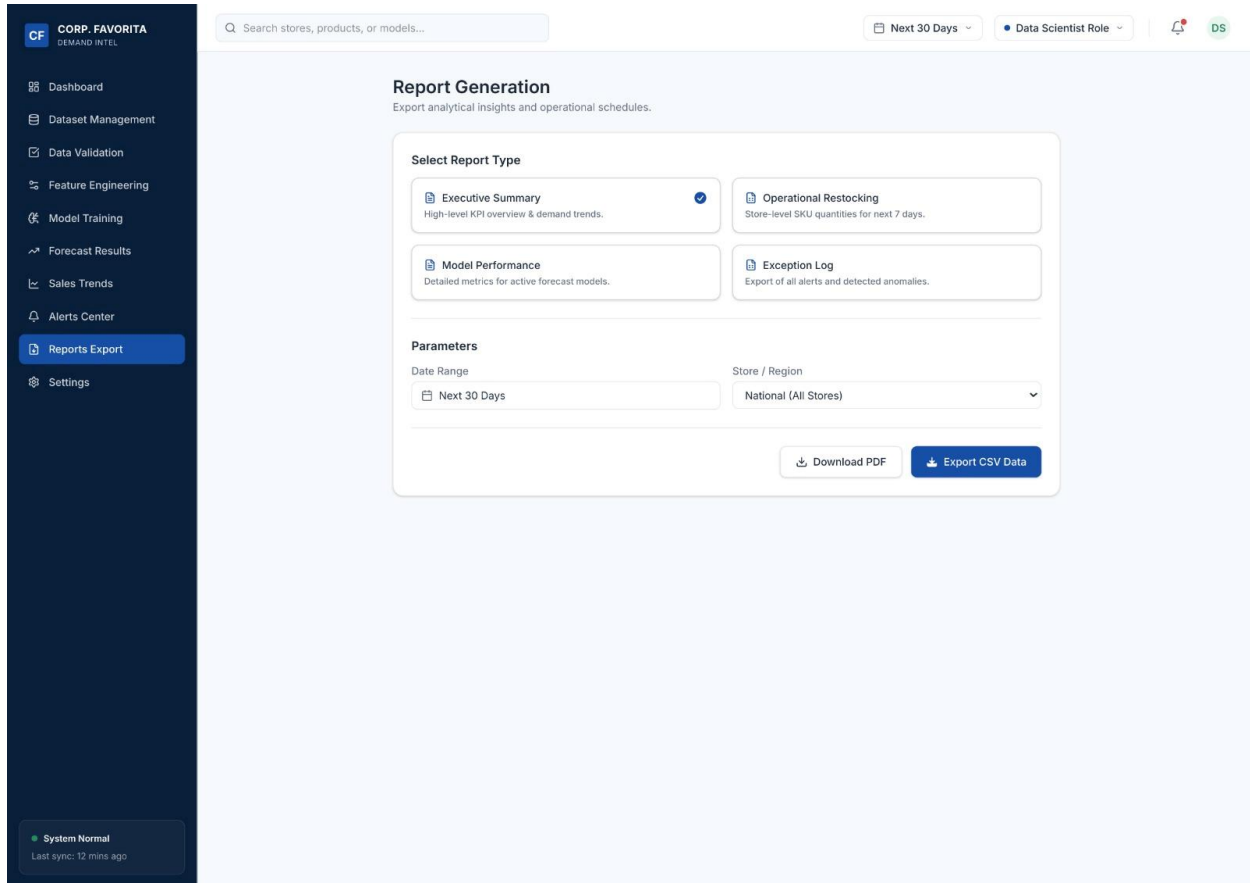
Monitor business anomalies and system health.

Mark All Read

All Alerts High Priority Inventory Risks ML Ops Resolved

- Imminent Stockout Risk** INVENTORY 10 mins ago  
Produce family at Store 44 is predicted to stockout in 48 hours.  
Investigate Acknowledge ALT-981
- Unexpected Demand Spike** SALES 2 hours ago  
Beverages seeing +25% unforecasted demand in coastal regions.  
Investigate Acknowledge ALT-892
- Model Performance Drop** ML OPS Yesterday, 14:30  
XGBoost Regressor accuracy dropped below 85% threshold.  
Mark Resolved ALT-885
- Data Pipeline Delayed** DATA ENG Oct 12, 02:45  
Daily Snowflake sync took longer than usual (45m vs avg 15m).  
ALT-874
- Retraining Completed** ML OPS Oct 11, 09:00  
Holiday Season Model finished training. Improved WAPE by 2%.  
ALT-878

**System Normal**  
Last sync: 12 mins ago

**Figure A17. Report Generation Mockup**

The mockup shows a web application interface for report generation. On the left is a dark blue sidebar with the company logo 'CF CORP. FAVORITA DEMAND INTEL' and a list of navigation items: Dashboard, Dataset Management, Data Validation, Feature Engineering, Model Training, Forecast Results, Sales Trends, Alerts Center, Reports Export (highlighted), and Settings. The main content area has a light blue background. At the top of this area is a search bar 'Search stores, products, or models...', a date range selector 'Next 30 Days', a role selector 'Data Scientist Role', and a user profile 'DS'. Below this is the 'Report Generation' section with the subtitle 'Export analytical insights and operational schedules.' It contains a 'Select Report Type' section with four options: 'Executive Summary' (selected), 'Operational Restocking', 'Model Performance', and 'Exception Log'. Each option has a brief description. Below this is a 'Parameters' section with a 'Date Range' dropdown set to 'Next 30 Days' and a 'Store / Region' dropdown set to 'National (All Stores)'. At the bottom right of the parameters section are two buttons: 'Download PDF' and 'Export CSV Data'. A system status bar at the bottom left of the sidebar shows 'System Normal' and 'Last sync: 12 mins ago'.

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Dashboard  
Dataset Management  
Data Validation  
Feature Engineering  
Model Training  
Forecast Results  
Sales Trends  
Alerts Center  
**Reports Export**  
Settings

Search stores, products, or models...

Next 30 Days Data Scientist Role DS

### Report Generation

Export analytical insights and operational schedules.

**Select Report Type**

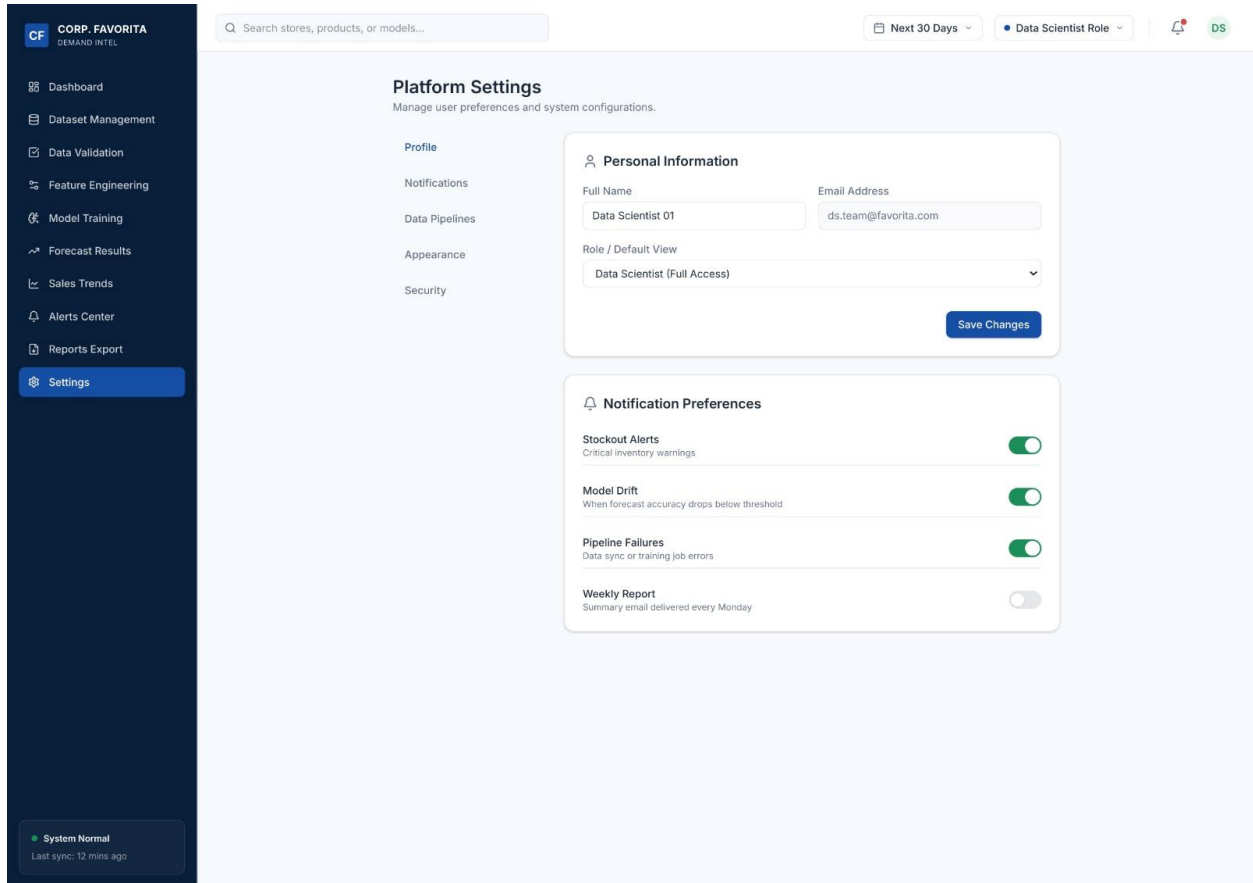
- Executive Summary**  
High-level KPI overview & demand trends.
- Operational Restocking**  
Store-level SKU quantities for next 7 days.
- Model Performance**  
Detailed metrics for active forecast models.
- Exception Log**  
Export of all alerts and detected anomalies.

**Parameters**

Date Range: Next 30 Days Store / Region: National (All Stores)

Download PDF Export CSV Data

System Normal  
Last sync: 12 mins ago

**Figure A18. Platform Settings Mockup**

The mockup shows a web application interface for 'CORP. FAVORITA DEMAND INTEL'. On the left is a dark blue sidebar with a list of navigation items: Dashboard, Dataset Management, Data Validation, Feature Engineering, Model Training, Forecast Results, Sales Trends, Alerts Center, Reports Export, and Settings (which is highlighted). At the bottom of the sidebar is a 'System Normal' status indicator. The main content area has a light blue background. At the top of this area is a search bar and two dropdown menus: 'Next 30 Days' and 'Data Scientist Role'. Below these is the 'Platform Settings' section, which includes a sub-header 'Manage user preferences and system configurations.' and a list of settings categories: Profile, Notifications, Data Pipelines, Appearance, and Security. The 'Profile' category is selected, showing a 'Personal Information' form with fields for Full Name (Data Scientist 01), Email Address (ds.team@favorita.com), and Role / Default View (Data Scientist (Full Access)). A 'Save Changes' button is at the bottom right of the form. Below the form is a 'Notification Preferences' section with four toggle switches: Stockout Alerts (on), Model Drift (on), Pipeline Failures (on), and Weekly Report (off).

**CORP. FAVORITA**  
DEMAND INTEL

Search stores, products, or models...

Next 30 Days Data Scientist Role

### Platform Settings

Manage user preferences and system configurations.

- Profile
- Notifications
- Data Pipelines
- Appearance
- Security

#### Personal Information

Full Name: Data Scientist 01

Email Address: ds.team@favorita.com

Role / Default View: Data Scientist (Full Access)

Save Changes

#### Notification Preferences

- Stockout Alerts** (On)  
Critical inventory warnings
- Model Drift** (On)  
When forecast accuracy drops below threshold
- Pipeline Failures** (On)  
Data sync or training job errors
- Weekly Report** (Off)  
Summary email delivered every Monday

System Normal  
Last sync: 12 mins ago

## **Appendix B. Source Notes**

- IBM Data Science Project - Round4.pdf: Project 5 requirements for Sales Forecasting and Demand Prediction.
- Project Documentation.pdf: documentation requirements for Project Planning & Management, Literature Review, Requirements Gathering, and System Analysis & Design.
- Architecture Style\_Machine Learning Pipeline Architecture.docx: project-specific machine learning pipeline architecture description.
- Software Architecture.drawio (2).pdf: software architecture diagram.
- Uploaded diagrams and UI/UX screenshots prepared for the project documentation.